Nonlinear Option Pricing

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The field of mathematics is very wide and it is not easy to predict what happens next, but I can tell you it is alive and well. Two general trends are obvious and will surely persist. In its pure aspect, the subject has changed, much for the better I think, by moving to more concrete problems. In both its pure and applied aspects, an equally beneficial shift to nonlinear problems can be seen. Most mathematical questions suggested by Nature are genuinely nonlinear, meaning very roughly that the result is not proportional to the cause, but varies with it as the square or the cube, or in some more complicated way. The study of such questions is still, after two or three hundred years, in its infancy. Only a few of the simplest examples are understood in any really satisfactory way. I believe this direction will be a principal theme in the future.

— **Henry P. McKean**, *Some Mathematical Coincidences (May 2003)*

Syllabus

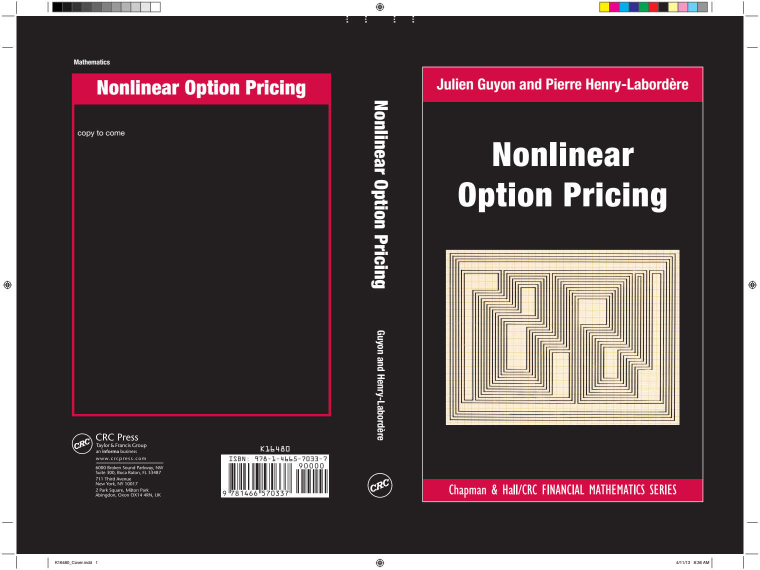
- The classical curriculum of mathematical finance programs generally covers the link between linear second order parabolic partial differential equations (PDEs) and stochastic differential equations (SDEs), resulting from Feynman-Kac's formula.
- However, many challenges faced by today's practitioners involve nonlinear PDEs.
- The aim of this course is to provide you with the mathematical tools and numerical methods required to tackle these issues, and illustrate the methods with practical case studies like:
  - American option pricing,
  - uncertain volatility, uncertain mortality, different rates for borrowing and lending,
  - calibration of models to market smiles,
  - credit valuation adjustment (CVA),
  - transaction costs, illiquid markets, super-replication under delta and gamma constraints, etc.

Syllabus

- We will spend some time on the theory:
  - optimal stopping,
  - stochastic control,
  - backward stochastic differential equations (BSDEs),
  - McKean SDEs and the particle method,
  - branching diffusions.
- But the main focus will deliberately be on ideas and numerical examples, which we believe help a lot in understanding the tools and building intuition.
- PDE methods suffer from the curse of dimensionality. Since most quantitative finance problems are high-dimensional, we will mostly focus on simulation-based methods (a.k.a. Monte Carlo algorithms).
- This course exposes the students with a wide variety of Machine Learning techniques, old and new, including parametric regression, nonparametric regression, neural networks, kernel trick, etc. These techniques allow us to compute some quantities (such as conditional expectations) that are key ingredients of the nonlinear Monte Carlo algorithms.

Syllabus

- Homeworks will allow you to check your understanding of the course by solving exercises inspired by our experience as quantitative analysts. You are expected to use the Python programming language to complete the homeworks.
- Final exam.
- If you have less than 10/20 at the final exam, you can get up to 3 more points by completing homeworks, but the final grade will not exceed 10.
- The main reference for this course is the monograph *Nonlinear Option Pricing* by Julien Guyon and Pierre Henry-Labordère.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Prerequisites

- We assume familiarity with classical stochastic analysis (Brownian motion, Itô formula, stochastic differential equations) and Black-Scholes option pricing.
- The students needing a reminder on these matters could consult:
  - Karatzas, I., Shreve, S.: Brownian Motion and Stochastic Calculus, Springer-Verlag, 1991.
  - Øksendal, B.: Stochastic Differential Equations: An Introduction with Applications, 5th ed., Springer-Verlag, 1998.
- We will cover linear option pricing: self-financing strategies, arbitrage, equivalent martingale measures (risk-neutral measures), market completeness, super-replication, Black-Scholes PDE, Feynman-Kac theorem (link between parabolic second order linear PDEs and SDEs).
- Regarding the new tools introduced, the course will be self-contained.

Numerical tools: Finite difference schemes vs Monte Carlo methods

Problems described by a nonlinear PDE

$$\begin{aligned} \partial_t u(t,x) + H(t,x,u(t,x),Du(t,x),D^2u(t,x)) &= 0, \quad x \in \Omega, \quad t \in [0,T), \\ u(T,x) &= g(x) \end{aligned} \quad (1)$$

- Ω: open domain of $\mathbb{R}^n$
- Du: gradient vector of the solution u with respect to spatial variables x
- D^2u: Hessian matrix of the solution u with respect to spatial variables x

In one dimension:

$$\begin{aligned} \partial_t u(t,x) + H(t,x,u(t,x),\partial_x u(t,x),\partial_x^2 u(t,x)) &= 0, \quad x \in \Omega, \quad t \in [0,T), \\ u(T,x) &= g(x) \end{aligned}$$

No analytical solution. Need for numerical approximation:

- Deterministic methods: finite difference schemes, finite elements
- Probabilistic methods (based on simulation): Monte Carlo

Finite difference schemes

- Localize the problem on a bounded domain
- Discretize the solution u on a space–time grid $\mathcal{G}_h = \Delta t\{0, 1, \ldots, n_T\} \times \Delta x\mathbb{Z}^n$ where $h = (\Delta t, \Delta x)$ and $n_T \Delta t = T$:

$$u_i^k = u(t_k, x_i), \qquad \forall (t_k, x_i) \in \mathcal{G}_h$$

- Replace the differentiation operators ∂_t, D, and D^2 by their finite difference approximations. In one dimension, one can use

$$\begin{aligned}
\partial_t^{\mathrm{fwd}} u(t_k, x_i) &= \frac{u_i^{k+1} - u_i^k}{\Delta t} \\
D^{\mathrm{fwd}} u(t_k, x_i) &= \frac{u_{i+1}^k - u_i^k}{\Delta x} \\
D^{\mathrm{bwd}} u(t_k, x_i) &= \frac{u_i^k - u_{i-1}^k}{\Delta x} \\
D^{\mathrm{cen}} u(t_k, x_i) &= \frac{u_{i+1}^k - u_{i-1}^k}{2\Delta x} \\
(D^2)^{\mathrm{cen}} u(t_k, x_i) &= \frac{u_{i+1}^k + u_{i-1}^k - 2u_i^k}{\Delta x^2}
\end{aligned}$$

Finite difference schemes

- PDE (1) can then be approximated by a (nonlinear) algebraic equation with unknowns u_i^k.
- By using centered discrete differentiation operators, an explicit scheme that uses centered finite differences reads

$$\frac{u_i^{k+1}-u_i^k}{\Delta t}+H(t_{k+1},x_i,u_i^{k+1},D^{\mathrm{cen}}u(t_{k+1},x_i),(D^2)^{\mathrm{cen}}u(t_{k+1},x_i))=0,$$
$$\forall (t_{k+1},x_i)\in\mathcal{G}_h$$

- The implicit scheme reads

$$\frac{u_i^{k+1}-u_i^k}{\Delta t}+H(t_k,x_i,u_i^k,D^{\mathrm{cen}}u(t_k,x_i),(D^2)^{\mathrm{cen}}u(t_k,x_i))=0,$$
$$\forall (t_k,x_i)\in\mathcal{G}_h\backslash\{t=T\}$$

- A θ-scheme reads

$$\frac{u_i^{k+1}-u_i^k}{\Delta t}+\theta H(t_{k+1},x_i,u_i^{k+1},D^{\mathrm{cen}}u(t_{k+1},x_i),(D^2)^{\mathrm{cen}}u(t_{k+1},x_i))$$
$$+(1-\theta)H(t_k,x_i,u_i^k,D^{\mathrm{cen}}u(t_k,x_i),(D^2)^{\mathrm{cen}}u(t_k,x_i))=0$$

Monte Carlo methods

- Curse of dimensionality: Finite diff schemes only work in small dimension
- No more than 3 space variables (assets and path-dependent variables)
- Need to turn to simulation-based methods (Monte Carlo methods)

Problem	Numerical tool
American options	ML techniques, duality
Chooser options	ML techniques, duality
Uncertain Lapse and Mortality	ML techniques, BSDEs
Uncertain Volatility	Parameterization, ML techniques, BSDEs
Different rates for borrowing/lending	ML techniques, BSDEs
Calibration of models to market smiles	Particle method, ML techniques, Malliavin
CVA	ML techniques, BSDEs, branching diffusion

ML techniques will mostly be used to estimate conditional expectations $\mathbb{E}[Y|X=x]$. They include: nearest neighbors, kernel regression, parametric regression, neural networks, random forests, kernel trick (+ cross-validation...). In this course we will apply them to simulated data, but they can be readily applied to data science problems.

Models of financial markets

- A filtered probability space $(\Omega, (\mathcal{F}_t)_{0\le t\le T}, \mathbb{P}^{\mathrm{hist}})$
- $\mathbb{P}^{\mathrm{hist}}$ is the historical or real probability measure under which we model our market.
- A market model is defined by an n-dimensional stochastic differential equation (SDE)

$$dX_t^i = b_i(t, X_t)\,dt + \sum_{j=1}^{d} \sigma_{i,j}(t, X_t)\,dW_t^j, \qquad i \in \{1, \ldots, n\} \tag{2}$$

and by another positive stochastic process B_t, called the money-market account, representing the value of cash, which satisfies

$$dB_t = r_t B_t\,dt, \qquad B_0 = 1$$

i.e.,

$$B_t = \exp\left(\int_0^t r_s\,ds\right)$$

- r_t is the short term interest rate.

Models of financial markets

- We set

$$D_{tu} := B_t B_u^{-1} = \exp\left(-\int_t^u r_s\, ds\right)$$

  which is the discount factor from date u to date t.
- Throughout the course, we will denote by $\tilde{Y}_t := D_{0t} Y_t$ the discounted value of any price process Y_t.
- Certain market components X^i may not be sold or bought in the market, such as the short term interest rate, or a stochastic volatility. Throughout this course, a market component X^i that can be sold and bought in the market is called an "asset."

Early Exercise Problems

Pricing and hedging of American options

If you want a happy ending, that depends, of course, on where you stop your story. — Orson Welles

European, American and Bermudan options

- A *European* option can be exercised only at the expiration date T,

$$V_t^E = \mathbb{E}^{\mathbb{Q}}\left[D_{t,T}F_T|\mathcal{F}_t\right]$$

- An *American* option may be exercised anytime on or before the expiration date T,

$$V_t^A = \sup_{\tau\in\mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}}\left[D_{t,\tau}F_\tau|\mathcal{F}_t\right]$$

  conditional on the option not being exercised at or before t, where $\mathcal{T}_{t,T}$ denotes the set of all stopping times τ such that $\tau \in [t,T]$.

- A *Bermudan* option can be exercised on certain pre-defined discrete set of dates $t_1 < \cdots < t_N = T$,

$$V_t^B = \sup_{\tau\in\mathcal{T}_D} \mathbb{E}^{\mathbb{Q}}\left[D_{t,\tau}F_\tau|\mathcal{F}_t\right]$$

  conditional on the option not being exercised at or before t, where $\mathcal{T}_D$ denotes the set of all stopping times $\tau \in \{t_1,\ldots,t_N\}\bigcap[t,T]$.

$$V_t^A \geq V_t^B \geq V_t^E$$

American Call and Put - No dividend

Assume positive interest rates. Martingale condition yields

$$\mathbb{E}^{\mathbb{Q}}\left[D_{t,T}S_T|\mathcal{F}_t\right] = S_t$$

American Call

Early exercise of an American call option leads to

- Loss of the time value of the option (from volatility)
- Loss of the time value of strike (from discounting)

Never optimal to exercise an American call option early.

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}\left[D_{t,T}\left(S_T-K\right)^+\middle|\mathcal{F}_t\right] &\geq \left(\mathbb{E}^{\mathbb{Q}}\left[D_{t,T}S_T-D_{t,T}K|\mathcal{F}_t\right]\right)^+\\ &= \left(S_t-D_{t,T}K\right)^+ \geq \left(S_t-K\right)^+\end{aligned}$$

American Put

Early exercise of an American put option leads to

- Loss of the time value of the option (from volatility)
- Gain on the time value of strike (from discounting)

Early exercise could be beneficial when the option is deeply in the money (time value of the option is small compared to the gain of the time value of the strike)

American Call and Put - Dividend

Continuous Dividend Yield $\mathbb{E}^{\mathbb{Q}}\left[D_{t,T}S_T|\mathcal{F}_t\right] = e^{-q(T-t)}S_t, \quad q > 0$

- American Call

  Early exercise of an American call option could be beneficial if the dividend yield is large enough compared to the time value of the option (from volatility) and the time value of strike (from discounting) lost

- American Put

  Large dividend yield benefits continuation

Discrete Dividend $S_{t_d-} - S_{t_d+} = D > 0, \quad D = \text{dividend}, \quad t_d = \text{ex-date}$

- American Call

  The optimal exercise time of an American call should be either right before the ex-date or at the final maturity

- American Put

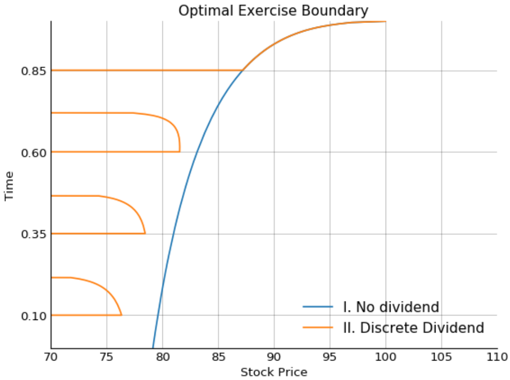
  It is never optimal to exercise an American put right before the ex-date

American Call and Put - Optimal Exercise Boundary

Example. American Put, $S = 100$, $K = 100$, $T = 1$, $\sigma = 25\%$, $r = 8\%$

I. no dividend

II. \$1 cash dividend paid at $t = 0.1, 0.35, 0.6, 0.85$



American Call and Put

- Put-Call parity does **not** hold for American options.
- Most listed options on individual stocks are American options with physical delivery
- Most listed options on stock indices are European options with cash settlement

Markovian Case - European options

Assume that there exists a (deterministic) function r such that $r_t = r(t, X_t)$ and g such that $F_T = g(X_T)$, then

$$u(t,x) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(s,X_s)ds} g(X_T) \middle| X_t = x \right]$$

Black-Scholes pricing PDE

$$\partial_t u(t,x) + \mathcal{L}u(t,x) - r(t,x)u(t,x) = 0, \quad u(T,x) = g(x).$$

where $\mathcal{L}$ is the Ito generator of X:

$$\mathcal{L} = \sum_{i=1}^{n} b_i(t,x)\partial_i + \frac{1}{2}\sum_{i,j=1}^{n}\sum_{k=1}^{d} \sigma_{i,k}(t,x)\sigma_{j,k}(t,x)\partial_{ij}$$

In dimension one $(n = d = 1)$,

$$\mathcal{L} = b(t,x)\frac{\partial}{\partial x} + \frac{1}{2}\sigma^2(t,x)\frac{\partial^2}{\partial x^2}$$

Markovian Case - American options

Consider an American option that pays $g(X_t)$ if exercised at time t.

Let $u(t,x)$ be the value of the American option at time t if $X_t = x$ and the option has not been exercised yet. Then

$$u(t,x) = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^\tau r(s,X_s)ds} g(X_\tau) \middle| X_t = x \right]$$

where $\mathcal{T}_{t,T}$ is the collection of all stopping times τ such that $t \leq \tau \leq T$.

Since $t \in \mathcal{T}_{t,T}$,

$$u(t,x) \geq g(x).$$

As long as $u(t,X_t) > g(X_t)$, it is not optimal to exercise, and the optimal stopping time τ^* is

$$\tau^* = \inf \left\{ s \geq t | u(s,X_s) = g(X_s) \right\}.$$

Markovian Case - American options - Variational Inequality

Variational Inequality

$$\max\left(\partial_t u(t,x) + \mathcal{L}u(t,x) - r(t,x)u(t,x),\; g(x) - u(t,x)\right) = 0$$
$$u(T,x) = g(x)$$

Let $\mathcal{J} = \partial_t + \mathcal{L} - r$.

- The option is worth at least its exercise value:

$$g(x) \leq u(t,x)$$

- The supermartingale property of $e^{-\int_0^t r(s,X_s)ds}u(t,X_t)$:

$$\mathcal{J}u(t,x) \leq 0$$

- $\mathcal{J}u = 0$ in the continuation region and $g = u$ in the exercise region:

$$\mathcal{J}u(t,x)\left(g(x) - u(t,x)\right) = 0$$

Markovian Case - American options - Variational Inequality

Formal proof of variational inequality

Assume that u is a smooth solution of the variational inequality

$$\max\left(\partial_t u(t,x) + \mathcal{L}u(t,x) - r(t,x)u(t,x),\; g(x) - u(t,x)\right) = 0$$

we shall show that u solves the optimal stopping problem, i.e.

$$u(t, X_t) = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}}\left[D_{t,\tau} g(X_t) | \mathcal{F}_t\right].$$

First, $\partial_t u + \mathcal{L}u - ru \leq 0$ implies that $D_{0,t}u(t, X_t)$ is a super-martingale so that for any $\tau \in \mathcal{T}_{t,T}$, by the optional sampling theorem and the fact $g(X_\tau) \leq u(\tau, X_\tau)$, we have

$$\mathbb{E}^{\mathbb{Q}}\left[D_{0,\tau} g\left(X_\tau\right) | \mathcal{F}_t\right] \leq \mathbb{E}^{\mathbb{Q}}\left[D_{0,\tau} u(\tau, X_\tau) | \mathcal{F}_t\right] \leq D_{0,t} u(t, X_t)$$

Taking the supremum over all $\tau \in \mathcal{T}_{t,T}$, we obtain

$$\sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}}\left[D_{t,\tau} g(X_\tau) | \mathcal{F}_t\right] \leq u(t, X_t).$$

Markovian Case - American options - Variational Inequality

Formal proof of variational inequality (continued)

On the other hand, define $\tau^* = \inf\{s \geq t | u(s, X_s) = g(X_s)\}$. Then

$$D_{0,\tau^*} u(\tau^*, X_{\tau^*}) = D_{0,t} u(t, X_t)$$
$$+ \int_t^{\tau^*} D_{0,s} \left(\partial_t u(s, X_s) + \mathcal{L}u(s, X_s) - r(s, X_s) u(s, X_s)\right) ds + \int_t^{\tau^*} \cdots dW_s$$

The definition of τ^* guarantees that $u(\tau^*, X_{\tau^*}) = g(X_{\tau^*})$ and

$$\partial_t u(s, X_s) + \mathcal{L}u(s, X_s) - r(s, X_s) u(s, X_s) = 0 \quad \text{for all } s \in [t, \tau^*],$$

so that

$$u(t, X_t) = \mathbb{E}^{\mathbb{Q}}\left[D_{t,\tau^*} g(X_{\tau^*}) | \mathcal{F}_t\right]$$

i.e. the supremum over $\mathcal{T}_{t,T}$ is attained by the optimal stopping time τ^*.

Markovian Case - Bermudan options

Assume the option can only be exercised on a selected number of dates $t_1, \ldots, t_N$:

$$u(t,x) = \sup_{\tau \in \mathcal{T}_{\{t_1,\ldots,t_N\} \cap [t,T]}} \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^\tau r(s,X_s)ds} g(X_\tau) \middle| X_t = x \right]$$

Between two dates t_{i+1} and t_i:

$$\partial_t u(t,x) + \mathcal{L}u(t,x) - r(t,x)u(t,x) = 0$$

At date t_i (**Jump condition**):

$$u(t_i^-, x) = \max(u(t_i^+, x), g(x))$$

Optimal stopping time

$$\tau^* = \inf\{s \in \{t_1, \ldots, t_N\} \cap [t,T] \mid u(s,X_s) = g(X_s)\}$$

Finite Difference Scheme

Finite difference scheme can be easily adapted to price options with early exercises features.

Consider a Bermudan option with exercise dates $t_1 < \cdots < t_N = T$ and payoff function $g(X_t)$

- Let $u(T, x) = g(x)$ and then proceed backward
- Between t_i and t_{i+1}, solve the PDE $\partial_t u + \mathcal{L}u - ru = 0$ by a classic finite difference scheme (fully explicit, fully implicit and Crank-Nicholson)
- Replace $u(t_i^+, x)$ by $u(t_i^-, x) = \max\left(u\left(t_i^+, x\right), g(x)\right)$ and continue the backward induction from t_i with the new terminal condition $u(t_i^-, x)$.

American options can be priced by applying the jump condition to every point in the time grid of the finite difference scheme. Usually only first-order in time.

Bermudan Options and Discrete Time Snell Envelope

Definition (Snell Envelope (Discrete Time))

Given a discrete-time stochastic process $\{Z_k\}_{1\leq k\leq N}$, the process U_k defined by

$$U_k = \sup_{\tau\in\mathcal{T}_{k,N}} \mathbb{E}\left[Z_\tau|\mathcal{F}_k\right], \quad k=1,\ldots,N$$

is called the *Snell envelope* of $\{Z_k\}$.

Consider a Bermudan option with exercise dates $t_1 < \cdots < t_N$ and payoffs $F_{t_1},\ldots,F_{t_N}$. Let V_{t_k} be the value of the option at time t_k if it has not been exercised yet.

$$\begin{aligned} V_{t_k} &= \sup_{\tau\in\mathcal{T}_{\{t_k,\ldots,t_N\}}} \mathbb{E}^{\mathbb{Q}}\left[D_{t_k,\tau}F_\tau|\mathcal{F}_{t_k}\right] \\ D_{0,t_k}V_{t_k} &= \sup_{\tau\in\mathcal{T}_{\{t_k,\ldots,t_N\}}} \mathbb{E}^{\mathbb{Q}}\left[D_{0,\tau}F_\tau|\mathcal{F}_{t_k}\right]. \end{aligned}$$

The discounted value process $\{D_{0,t_k}V_{t_k}\}_{k=1}^N$ is the Snell envelope of the discounted payoff $\{D_{0,t_k}F_{t_k}\}_{k=1}^N$.

Recursive Scheme of Building Discrete Time Snell Envelope

The Snell envelope $\{U_k\}$ can be constructed explicitly by the following recursive scheme:

$$U_k = \sup_{\tau \in \mathcal{T}_{k,N}} \mathbb{E}\left[Z_\tau | \mathcal{F}_k\right], \quad k = 0, \dots, N \tag{3}$$

if and only if

$$\begin{cases} U_N = Z_N \\ U_k = \max\{Z_k, \mathbb{E}\left[U_{k+1} | \mathcal{F}_k\right]\} \quad k = 0, \dots, N-1 \end{cases} \tag{4}$$

Proof. First assume U_k is given by (3). Obviously, $U_N = Z_N$. Let $\tau^*_{k+1} = \text{argmax}_{\tau \in \mathcal{T}_{k+1,N}} \mathbb{E}\left[Z_\tau | \mathcal{F}_{k+1}\right]$ be the optimal stopping time. Then

$$\begin{aligned} \mathbb{E}\left[U_{k+1} | \mathcal{F}_k\right] &= \mathbb{E}\left[\mathbb{E}\left[Z_{\tau^*_{k+1}} \middle| \mathcal{F}_{k+1}\right] \middle| \mathcal{F}_k\right] \\ &= \mathbb{E}\left[Z_{\tau^*_{k+1}} \middle| \mathcal{F}_k\right] \leq \sup_{\tau \in \mathcal{T}_{k,N}} \mathbb{E}\left[Z_\tau | \mathcal{F}_k\right] = U_k \end{aligned}$$

The inequality holds since $\tau^*_{k+1} \in \mathcal{T}_{k,N}$. Obviously, $Z_k \leq U_k$ since $k \in \mathcal{T}_{k,N}$. Hence $\max\{Z_k, \mathbb{E}\left[U_{k+1} | \mathcal{F}_k\right]\} \leq U_k$.

On the other hand, for any $\tau \in \mathcal{T}_{k,N}$, $Z_\tau = Z_k \mathbf{1}_{\{\tau = k\}} + Z_{\tau \vee (k+1)} \mathbf{1}_{\{\tau > k\}}$.

Recursive Scheme of Building Discrete Time Snell Envelope

$$\begin{aligned}
\mathbb{E}\left[Z_\tau|\mathcal{F}_k\right] &= Z_k\mathbf{1}_{\{\tau=k\}} + \mathbb{E}\left[Z_{\tau\vee(k+1)}|\mathcal{F}_k\right]\mathbf{1}_{\{\tau>k\}} \\
&= Z_k\mathbf{1}_{\{\tau=k\}} + \mathbb{E}\left[\mathbb{E}\left[Z_{\tau\vee(k+1)}|\mathcal{F}_{k+1}\right]|\mathcal{F}_k\right]\mathbf{1}_{\{\tau>k\}} \\
&\leq Z_k\mathbf{1}_{\{\tau=k\}} + \mathbb{E}\left[U_{k+1}|\mathcal{F}_k\right]\mathbf{1}_{\{\tau>k\}} \\
&\leq \max\{Z_k, \mathbb{E}\left[U_{k+1}|\mathcal{F}_k\right]\}
\end{aligned}$$

Thus

$$U_k = \sup_{\tau\in\mathcal{T}_{k,N}} \mathbb{E}\left[Z_\tau|\mathcal{F}_k\right] \leq \max\{Z_k, \mathbb{E}\left[U_{k+1}|\mathcal{F}_k\right]\}.$$

Next assume that U_k is given by the recursive scheme (4). Clearly U_k is a supermartingale that dominates Z_k. Let $\tau_k^* = \inf\{n \geq k; Z_n = U_n\}$. Then

$$U_{(n+1)\wedge\tau_k^*} - U_{n\wedge\tau_k^*} = 1_{n+1\leq\tau_k^*}\left(U_{n+1} - U_n\right) = 1_{n+1\leq\tau_k^*}\left(U_{n+1} - \mathbb{E}\left[U_{n+1}|\mathcal{F}_n\right]\right)$$

Note that $\{n+1 \leq \tau_k^*\} \in \mathcal{F}_n$. Hence $\mathbb{E}\left[U_{(n+1)\wedge\tau_k^*}\middle|\mathcal{F}_n\right] = U_{n\wedge\tau_k^*}$, i.e. $U_{n\wedge\tau_k^*}$ is a martingale so that $U_k = \mathbb{E}\left[U_{\tau_k^*}\middle|\mathcal{F}_k\right] = \mathbb{E}\left[Z_{\tau_k^*}\middle|\mathcal{F}_k\right]$. Furthermore, for $\forall\tau\in\mathcal{T}_{k,N}$,

$$\mathbb{E}\left[Z_\tau|\mathcal{F}_k\right] \leq \mathbb{E}\left[U_\tau|\mathcal{F}_k\right] \leq U_k.$$

$\square$

Discrete-Time Snell Envelope and Bermudan Options

Corollary. Given $\{Z_n\}_{n\leq N}$, the Snell envelope $\{U_k\}$ is the smallest supermartingale dominating $\{Z_n\}$. For $\forall k \leq N$, the optimal stopping time τ_k^* is given by

$$\tau_k^* = \inf\{n \geq k; Z_n = U_n\}$$

and the stopped process $\{U_{n\wedge\tau_k^*}\}_{n\geq k}$ is a martingale.

Bermudan Option

Let V_{t_k} be the price of a Bermudan option at time t_k provided the option has not been exercised before t_k. Then $\{D_{0,t_k}V_{t_k}\}$ is the smallest supermartingale that dominates the discounted payoff $\{D_{0,t_k}F_{t_k}\}$ and can be constructed by the recursive scheme:

$$\begin{cases} V_{t_N} & = & F_{t_N} \\ V_{t_i} & = & \max\left(F_{t_i}, \mathbb{E}^{\mathbb{Q}}[D_{t_i t_{i+1}} V_{t_{i+1}} | \mathcal{F}_{t_i}]\right) \end{cases}$$

If the option has not been exercised before t_k, the optimal stopping time τ_k^* is

$$\tau_k^* = \inf\{t_j \geq t_k \,|\, V_{t_j} = F_{t_j}\}$$

The stopped value process $\{D_{0,t_j\wedge\tau_k^*}V_{t_j\wedge\tau_k^*}\}_{j\geq k}$ is a martingale.

Monte Carlo Methods

Why do we need it?

Curse of dimensionality for finite difference methods when the number of variables exceeds three.

- Multi-asset options
- Path-dependent payoff
- Stochastic volatility/stochastic interest rates

Why is it difficult?

Need to estimate the continuation value $\mathbb{E}^{\mathbb{Q}}\left[D_{t_i,t_{i+1}}V_{i+1}\middle|\mathcal{F}_{t_i}\right]$. Naive nested Monte Carlo is explosive.

Backward Induction

First, we approximate American options by Bermudan options with exercise dates $t_1, t_2, \dots, t_N = T$ and payoff $F_{t_1}, F_{t_2}, \dots, F_{t_N}$.

Let V_{t_i} be the value of the option at time t_i.

At the final expiry $t_N = T$,

$$V_{t_N} = F_{t_N}$$

Then we proceed by backward induction

$$V_{t_i} = \max\left(\mathbb{E}^{\mathbb{Q}}\left[D_{t_i,t_{i+1}} V_{t_{i+1}} | \mathcal{F}_{t_i}\right], g(X_{t_i})\right), \quad i = 1, \dots, N-1$$

The key is to determine the continuation value $C_{t_i} = \mathbb{E}^{Q}\left[D_{t_i,t_{i+1}} V_{t_{i+1}} | \mathcal{F}_{t_i}\right]$

Regression algorithms

Continuation value at t_i:

$$C_{t_i} = \mathbb{E}^{\mathbb{Q}}\left[D_{t_i,t_{i+1}} V_{t_{i+1}} | \mathcal{F}_{t_i}\right] \quad \text{(A)}$$

Let τ_{i+1} be the optimal stopping time at t_{i+1}. Then

$$V_{t_{i+1}} = \mathbb{E}^{\mathbb{Q}}\left[D_{t_{i+1},\tau_{i+1}} F_{\tau_{i+1}} | \mathcal{F}_{t_{i+1}}\right]$$

so that

$$C_{t_i} = \mathbb{E}^{\mathbb{Q}}\left[D_{t_i,\tau_{i+1}} F_{\tau_{i+1}} | \mathcal{F}_{t_i}\right] \quad \text{(B)}$$

The conditional expectation can be estimated from the cross-sectional information in the simulated paths.

- Tsitsiklis-Van Roy (2001) uses (A) and regress $\hat{V}(t_{i+1}, X_{t_{i+1}})$ on some functions of X_{t_i}
- Longstaff-Schwartz (2001) uses (B) and regress the sum of subsequent realized cash flows on some functions of X_{t_i} (more robust, more accurate)

The Tsitsiklis-Van Roy algorithm

Bermudan option with exercise dates $t_1 < \cdots < t_N$ and payoff $F_{t_1}, \ldots, F_{t_N}$. Let $\hat{V}_{t_i}(\omega)$ denote the approximation to $V_{t_i}(\omega)$ on each path.

Procedure

- Simulate paths until maturity $T = t_N$
- Let $\hat{V}_{t_N} = F_{t_N}$
- For $i = N-1, N-2, \ldots, 1$, estimate the continuation value

$$\hat{C}_{t_i} = \mathbb{E}^{\mathbb{Q}}[D_{t_i t_{i+1}} \hat{V}_{t_{i+1}} | \mathcal{F}_{t_i}]$$

  and then let

$$\hat{V}_{t_i} = \max(F_{t_i}, \hat{C}_{t_i})$$

- Estimate V_0 by

$$\hat{V}_0 = \mathbb{E}^{\mathbb{Q}}[D_{0t_1} \hat{V}_{t_1}]$$

  where $\mathbb{E}^{\mathbb{Q}}$ is replaced by the average value over all simulated paths

The Longstaff-Schwartz algorithm

Procedure

- Simulate paths until maturity $T = t_N$
- Let $\hat{\tau}_N = t_N$
- For $i = N-1, \dots, 1$,

$$\hat{C}_{t_i} = \mathbb{E}^{\mathbb{Q}}[D_{t_i \hat{\tau}_{i+1}} F_{\hat{\tau}_{i+1}} | \mathcal{F}_{t_i}]$$

$$\hat{\tau}_i = \begin{cases} t_i & \text{if} \quad \hat{C}_{t_i} \leq F_{t_i} \\ \hat{\tau}_{i+1} & \text{otherwise} \end{cases}$$

- Eventually, estimate the optimal strategy by $\hat{\tau} = \hat{\tau}_1$ and the price V_0 by

$$\hat{V}_0 = \mathbb{E}^{\mathbb{Q}}[D_{0\hat{\tau}} F_{\hat{\tau}}]$$

where $\mathbb{E}^{\mathbb{Q}}$ is replaced by the average value over all simulated paths

Tsitsiklis-Van Roy v.s. Longstaff-Schwartz

Both Tsitsiklis-Van Roy (TVR) and Longstaff-Schwartz (LS) estimate the continuation value from the cross-sectional information in the simulated paths by regression.

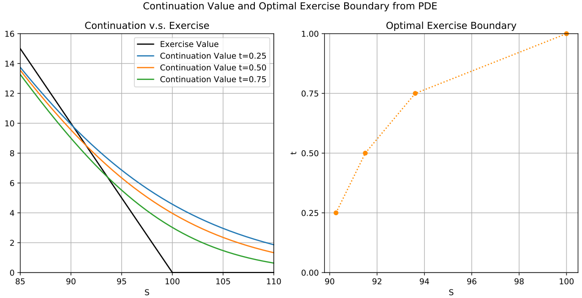
- TVR uses the fitted values from the previous step in the regression at current step whereas LS uses subsequent realized cash flows.
- The continuation estimate $\hat{C}_{t_i}$ is used directly in building value function V_{t_i} in TVR, but only used in making exercise decision in LS.
- The regression error from estimating $\hat{C}_{t_i}$ leads to error in $\hat{V}_{t_i}$ in TVR, but error in exercise decision in LS. In both algorithms the errors propagate through the backward induction. In general the error propagation is worse in TVR.
- In LS, one only needs to estimate the continuation value where option is in the money.
- In general, the LS algorithm is more accurate and robust than the TVR algorithm.

Tsitsiklis-Van Roy v.s. Longstaff-Schwartz - Example

Example: Bermudan put option, quarterly exercise, Black-Scholes model,

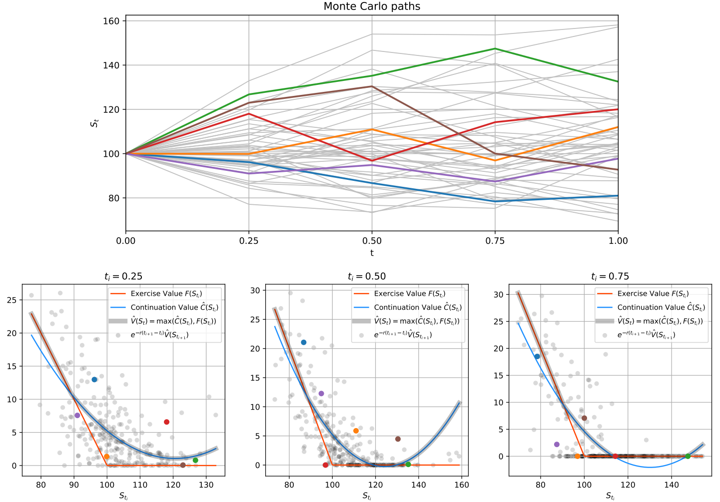
$$S_0 = 100, \sigma = 0.2, r = 0.1, q = 0.02, K = 100, T = 1. \quad F(S) = (K - S)^+.$$

PDE:

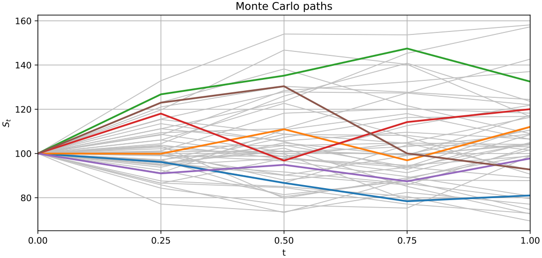


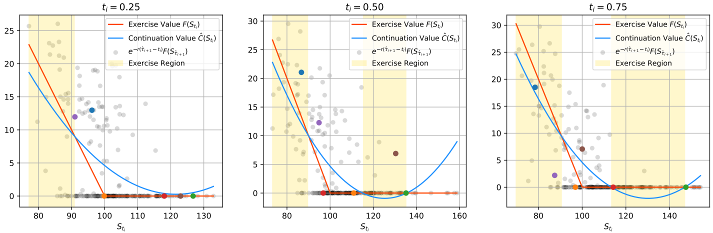
Monte Carlo: estimate of continuation value is obtained by quadratic polynomial fit

Tsitsiklis-Van Roy - Example



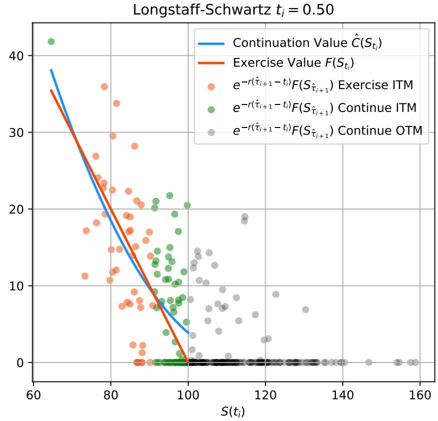
Longstaff-Schwartz - Example





Longstaff-Schwartz - Example

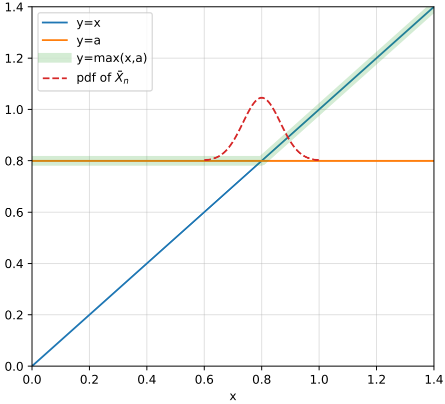
Continuation values can be estimated only where the option is in the money.



Convexity of $\max(x, a)$ and Jensen's Bias

Example. Let $X_1, \ldots, X_n$ be independent samples of a random variable X and $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample mean. Assume that $\mathbb{E}[X] = a = 0.8$. Then

$$\mathbb{E}\left[\max\left(a, \bar{X}_n\right)\right] \geq \max(a, \mathbb{E}[X]) = a.$$



Bias and Lower Bound Pricing

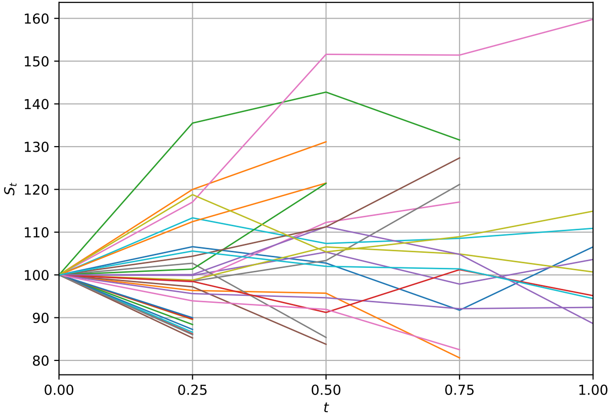
- There are both bias and variance from the estimation of the continuation value $\hat{C}_{t_i}$, which lead to bias in the estimation of the final price $\hat{V}_0$.
- The variance in $\hat{C}_{t_i}$ creates high bias in $\hat{V}_0$ due to Jensen's bias. The Jensen's bias diminishes as the number of paths goes to infinity.
- The bias in $\hat{C}_{t_i}$ typically leads to the bias of the same direction in $\hat{V}_0$ in the TVR algorithm.
- The bias in $\hat{C}_{t_i}$ leads to a sub-optimal strategy, which produces low bias in $\hat{V}_0$ in the Longstaff-Schwartz algorithm.
- Typically, both Tsitsiklis-Van Roy and Longstaff-Schwartz algorithms have undetermined bias.
- *Lower Bound (or Low-biased) pricing* requires a two-step procedure.
  1. Run TVR or Longstaff-Schwartz algorithm with p_1 paths and obtain an estimate of the optimal strategy $\hat{\tau}_1$:

     $$\hat{\tau}_1 = \inf_{j=1,\dots,N}\{t_j | \hat{C}_{t_j} <= F_{t_j}\}$$

  2. Simulate p_2 new independent paths that are stopped according to $\hat{\tau}_1$.

Lower Bound Pricing - Example

Example: Independent Monte Carlo simulation to get lower bound price with exercise strategy given by the Longstaff-Schwartz algorithm.



Conditional Expectation $\mathbb{E}\left[Y|X\right]$

1. (Best Predictor). If $Y \in L^2$, $f^*(X) \triangleq \mathbb{E}\left[Y|X\right]$ minimizes the mean squared error $\mathbb{E}\left[(Y - f(X))^2\right]$ among all f s.t. $f(X) \in L^2$. Therefore $f^*(X)$ can be interpreted as a projection of Y on the space of all functions of X and is the best predictor of Y (with the smallest mean square error) given the value of X.

2. If both X and Y are continuous random variables, then

$$\mathbb{E}\left[Y|X = x\right] = \int y f_{Y|X}(y|x) dy$$

where the conditional density $f_{Y|X}$ is the ratio of the joint density $f_{X,Y}$ and marginal density f_X,

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

Conditional Expectation Estimation - Parametric Regression

We approximate $\mathbb{E}[Y|X]$ by restricting the subspace of functions of X to a smaller subspace of functions of some parametric form $\{f(x;\alpha), \alpha \in \mathcal{A}\}$:

$$\alpha^* = \text{argmin}_{\alpha \in \mathcal{A}} \mathbb{E}\left[(Y - f(X;\alpha))^2\right] \quad \text{and} \quad \mathbb{E}[Y|X] \approx f(X;\alpha^*)$$

In particular, we may choose $f(x;\alpha) = \sum_{k=1}^n \alpha_k f_k(x)$ to be the linear sum of basis functions $\{f_k\}_{k=1}^n$, i.e.

$$Y \approx \sum_{k=1}^n \alpha_k f_k(X) + \epsilon \quad \text{where } \epsilon \text{ is independent noise and} \quad \mathbb{E}[Y|X] \approx \sum_{k=1}^n \alpha_k f_k(X)$$

Given N observations $((x_1, y_1), \ldots, (x_N, y_N))$ of X and Y, finding $\mathbb{E}[Y|X]$ comes down to solving least square problem $\min_{\alpha \in \mathbb{R}^n} \|A\alpha - y\|_2$, where

$$A = \begin{bmatrix} f_1(x_1) & \cdots & f_n(x_1) \\ \vdots & \ddots & \vdots \\ f_1(x_N) & \cdots & f_n(x_N) \end{bmatrix} \quad \text{and} \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

$$\alpha^* = \left(A^T A\right)^{-1} A^T y.$$

Conditional Expectation Estimation - Parametric Regression

- Select simple regressors and avoid over fitting problem.
- Select regressors similar to the payoff function.
- Piecewise polynomials can be used as basis functions, e.g., piecewise-linear functions, regression splines.
- Use `numpy.polyfit(x, y)` for polynomial regression and `numpy.linalg.lstsq(A, b)` for general parametric regression.
- Avoid overfitting with regularization (Ridge Regression or Tikhonov regularization):

$$\min_{\alpha \in \mathbb{R}^n} \|A\alpha - y\|_2^2 + \lambda \|\alpha\|_2^2, \quad \lambda > 0$$

  - λ is a hyperparameter (remaining fixed during optimization/training). The optimal solution α^* is given by

$$\alpha^* = \left(A^T A + \lambda I\right)^{-1} A^T y.$$

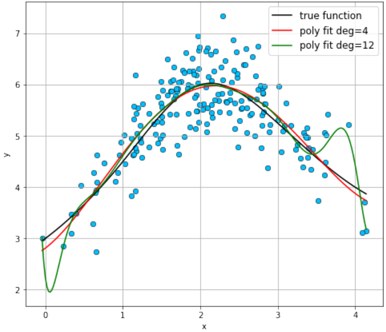
  - Normalize variables beforehand if necessary
  - Penalty can be applied to selected variables
  - Use `sklearn.linear_model.Ridge` for ridge regression

Conditional Expectation Estimation - Polynomial Regression

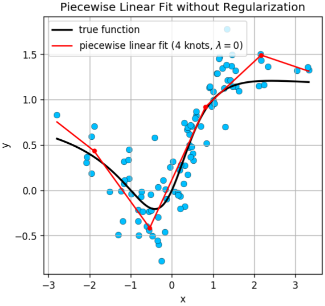
Polynomial basis:

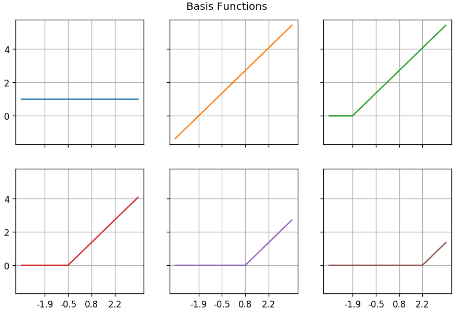
$$f_1(x) = 1, \quad f_2(x) = x, \quad f_3(x) = x^2, \; \ldots, \; f_{n+1}(x) = x^n$$

Polynomial regression with higher order polynomials are more likely to overfit; Its behavior near the boundaries tends to be erratic and extrapolation can be dangerous.



Conditional Expectation Estimation - Piecewise Linear Fit





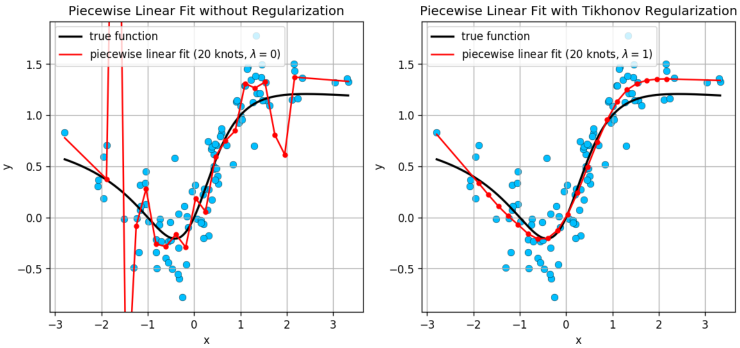
For given knots $x_1 < \cdots < x_n$, the basis functions are

$$f_1(x) = 1, \ \ f_2(x) = x - x_1$$

$$f_{2+i}(x) = (x - x_i)_+, \quad i = 1, \ldots, n$$

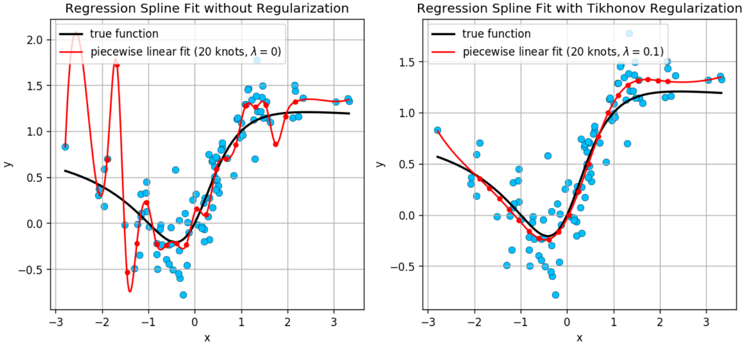
Conditional Expectation Estimation - Piecewise Linear Fit

Tikhonov Regularization



Penalties are applied to α_{2+i}, $i = 1, \ldots, n$.

Conditional Expectation Estimation - Regression Cubic Splines



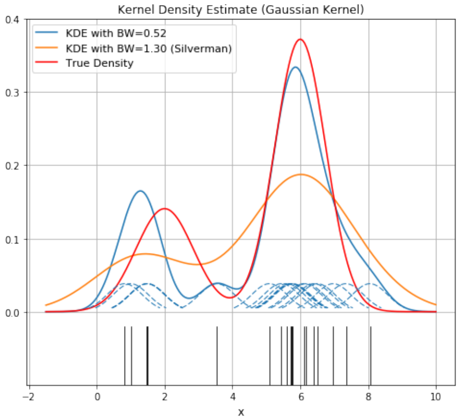
For given knots $x_1 < \cdots < x_n$, the basis functions are

$$f_1(x) = 1,\ f_2(x) = x - x_1,\ f_3(x) = (x - x_1)^2,\ f_4(x) = (x - x_1)^3$$

$$f_{4+i}(x) = (x - x_i)_+^3, \quad i = 1, \ldots, n$$

Penalties are applied to α_{4+i}, $i = 1, \ldots, n$.

Kernel Density Estimation (KDE)



$$\hat{f}_h(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x-x_i}{h}\right) \quad \text{with} \quad K \geq 0 \text{ and } \int K(x)dx = 1$$

Kernel Density Estimation (KDE)

Commonly used kernel functions:

- Epanechnikov $K(x) = \frac{3}{4}\left(1 - x^2\right)$ for $-1 \leq x \leq 1$ and 0 elsewhere;
- Quartic $K(x) = \frac{15}{16}(1+x)^2(1-x)^2$ for $-1 \leq x \leq 1$ and 0 elsewhere;
- Gaussian $K(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ for all x.

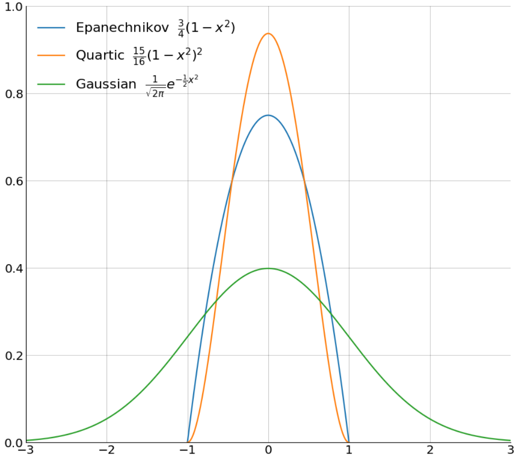
Bandwidth Selection:

Balance the bias-variance tradeoff. Large h gives lower variance but higher bias. Silverman's rule of thumb:

$$h = \left(\frac{4\hat{\sigma}^5}{3N}\right)^{\frac{1}{5}} \approx 1.06\hat{\sigma}N^{-\frac{1}{5}} \quad \text{where } \hat{\sigma}^2 \text{ is the sample variance}$$

Optimal (in integrated mean square error terms) if Gaussian kernel is used to estimate Gaussian density.

Kernel Density Estimation (KDE) - Commonly Used Kernels



Conditional Expectation Estimation - Nonparametric Regression

Nadaraya-Watson Kernel Regression (Locally weighted average)

$$\mathbb{E}[Y|X=x] \approx \frac{\sum_{i=1}^{N} K_h(x-x_i) y_i}{\sum_{i=1}^{N} K_h(x-x_i)}$$

where K is the kernel, $h > 0$ is the bandwidth and $K_h(x) = K(x/h)/h$.

Derivation.

$$\mathbb{E}\left[Y|X=x\right] = \int y \frac{f_{X,Y}(x,y)}{f_X(x)} dy$$

Use kernel density estimation for f_X and $f_{X,Y}$ with kernel K_h:

$$f_X(x) \approx \frac{1}{n} \sum_{i=1}^{n} K_h(x-x_i), \quad f_{X,Y}(x,y) \approx \frac{1}{n} \sum_{i=1}^{n} K_h(x-x_i) K_h(y-y_i)$$

and (assuming that K is centered: $\int y K_h(y) dy = 0$) the fact

$$\int y K_h(y-y_i) dy = y_i.$$

$\square$

Conditional Expectation Estimation - Nonparametric Regression

Local Linear Regression

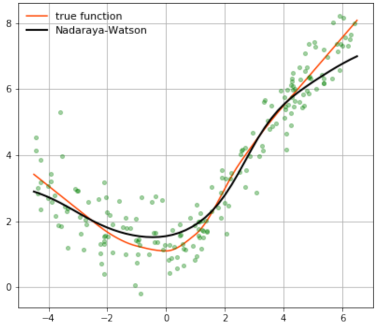
The locally weighted linear regression solves a separate weighted least squares problem at each target point x,

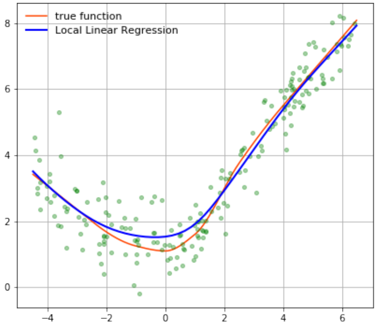
$$\min_{\alpha,\beta} \sum_{i=1}^{N} K_h(x - x_i) \left[y_i - \alpha - \beta x_i\right]^2$$

Then the estimate is $\hat{\alpha} + \hat{\beta} x$, where $\hat{\alpha}$ and $\hat{\beta}$ are dependent on x.

- The locally-weighted averages can be badly biased on the boundaries. This bias can be removed by local linear regression to first order.
- Cross validation can be used for bandwidth selection.

NW v.s. Local Linear Regression



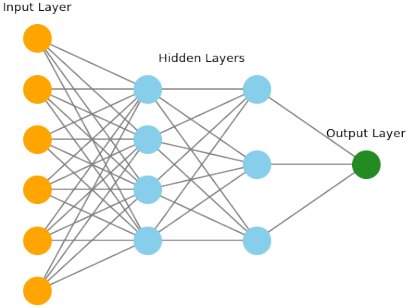


Artificial Neural Networks

Feedforward Neural Network

Inputs:	$a_0 = x \in \mathbb{R}^p$
Hidden layers	$a_i = f_i\left(W_i a_{i-1} + b_i\right), \quad i = 1, \dots, N-1$
Outputs	$y = a_N = W_N a_{N-1} + b_N \in \mathbb{R}^q$

The nonlinear function f_i is called activation function. Some common choices are sigmoid, hyperbolic tangent, ReLu and ELU functions.



Universal Approximation Theorem

Theorem

Let ϕ be a non-constant bounded and monotone-increasing continuous function on $\mathbb{R}$. Given any continuous function f on $[0,1]^n$ and $\epsilon > 0$, there exists an integer m and real constants β_i, b_i and w_{ij}, with $i = 1, \ldots, m$ and $j = 1, \ldots, n$, such that the function F defined by

$$F(x_1, \ldots, x_n) = \sum_{i=1}^{m} \beta_i \phi \left(\sum_{j=1}^{n} w_{ij} x_j + b_i \right)$$

is a uniform ϵ-approximation to f:

$$|F(x_1, \ldots, x_n) - f(x_1, \ldots, x_n)| < \epsilon \quad \text{for all } x_1, \ldots, x_n.$$

- Any continuous function can be approximated to arbitrary accuracy by a neural network with a single hidden layer, provided the hidden layer is sufficiently wide.

Artificial Neural Networks

- A neural network *learns* the basis functions instead of using fixed ones.
- Much of the power of neural network comes from the repeated composition of simpler functions to represent a more complex function.
- A loss function needs to be specified to train the model. For regression tasks, a typical choice is the mean squared error.
- Gradient with respect to the network parameters can be computed efficiently by *backpropagation*. Then the network parameters can be updated by a gradient-based optimizer such as stochastic gradient descent.
- A neural network typically has a huge number of parameters. Regularization techniques are helpful to avoid overfitting.
- The network parameters are initialized randomly (with variances properly chosen) to break the symmetry and avoid the exploding/vanishing gradient problems.
- It is also important to fine-tune the hyperparameters.
- Popular deep learning libraries include TensorFlow, CNTK, Theano and PyTorch. Keras is a high-level API with multibackend support.

Primal and Dual Problems

Primal Problem (maximizing over stopping times)

$$\sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}} \left[D_{t,\tau} F_\tau | \mathcal{F}_t \right] \tag{5}$$

Dual Problem (minimizing over martingales) [Rogers 2002; Haugh & Kogan 2004]

$$\inf_{M \in \mathcal{M}_{t,0}} \mathbb{E}^{\mathbb{Q}} \left[\sup_{t \leq s \leq T} \left(D_{t,s} F_s - M_s \right) \middle| \mathcal{F}_t \right] \tag{6}$$

where $\mathcal{M}_{t,0}$ denotes the set of all right-continuous martingales $(M_s)_{s \in [t,T]}$ with $M_t = 0$. Let $(U_s)_{s \in [t,T]}$ be the Snell envelope of discounted payoff $(D_{t,s} F_s)_{s \in [t,T]}$. Then the optimal martingale M^* is the martingale part of the Doob-Meyer decomposition of $(U_s)_{s \in [t,T]}$, minus its value at time t.

$$\boxed{\sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}} \left[D_{t,\tau} F_\tau | \mathcal{F}_t \right] = \inf_{M \in \mathcal{M}_{t,0}} \mathbb{E}^{\mathbb{Q}} \left[\sup_{t \leq s \leq T} \left(D_{t,s} F_s - M_s \right) \middle| \mathcal{F}_t \right]}$$

Primal and Dual Problems

Proof. For any $\tau \in \mathcal{T}_{t,T}$ and $M \in \mathcal{M}_{t,0}$, it follows from the optional sampling theorem that

$$\mathbb{E}^{\mathbb{Q}}\left[M_\tau | \mathcal{F}_t\right] = M_t = 0$$

Therefore

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}\left[D_{t,\tau} F_\tau | \mathcal{F}_t\right] &= \mathbb{E}^{\mathbb{Q}}\left[D_{t,\tau} F_\tau - M_\tau | \mathcal{F}_t\right] + \mathbb{E}^{\mathbb{Q}}\left[M_\tau | \mathcal{F}_t\right] \\ &\leq \mathbb{E}^{\mathbb{Q}}\left[\sup_{t \leq s \leq T} \left(D_{t,s} F_s - M_s\right) \middle| \mathcal{F}_t\right]\end{aligned}$$

so that

$$\sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}}\left[D_{t,\tau} F_\tau | \mathcal{F}_t\right] \leq \inf_{M \in \mathcal{M}_{t,0}} \mathbb{E}^{\mathbb{Q}}\left[\sup_{t \leq s \leq T} \left(D_{t,s} F_s - M_s\right) \middle| \mathcal{F}_t\right]. \tag{7}$$

On the other hand, for any fixed t, let $U_s = \sup_{\tau \in \mathcal{T}_{s,T}} \mathbb{E}^{\mathbb{Q}}\left[D_{t,\tau} F_\tau | \mathcal{F}_s\right]$, $s \in [t,T]$, be the Snell envelope of the discounted payoff $\left(D_{t,s} F_s\right)_{s \in [t,T]}$. Then $\left(U_s\right)_{s \in [t,T]}$ is the smallest supermartingale that dominates $\left(D_{t,s} F_s\right)_{s \in [t,T]}$. In particular, it has unique Doob-Meyer decomposition

$$U_s = M_s - A_s, \quad \forall s \in [t,T]$$

where $\left(M_s\right)_{s \in [t,T]}$ is a martingale with $M_t = U_t$ and $\left(A_s\right)_{s \in [t,T]}$ a predictable increasing process with $A_t = 0$.

Primal and Dual Problems

Let $M_s^* = M_s - M_t$, $s \in [t,T]$. Then

$$D_{t,s}F_s - M_s^* = D_{t,s}F_s - U_s - A_s + M_t \leq M_t = U_t.$$

Since this holds for all $s \in [t,T]$, we must have

$$\sup_{t \leq s \leq T} (D_{t,s}F_s - M_s^*) \leq U_t = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}} \left[D_{t,\tau}F_\tau | \mathcal{F}_t \right].$$

Taking conditional expectation $\mathbb{E}^{\mathbb{Q}}[\cdot|\mathcal{F}_t]$ and using inequality (7) we obtain

$$\mathbb{E}^{\mathbb{Q}} \left[\sup_{t \leq s \leq T} (D_{t,s}F_s - M_s^*) \middle| \mathcal{F}_t \right] = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}} \left[D_{t,\tau}F_\tau | \mathcal{F}_t \right] \qquad (8)$$

which shows that the infimum in (6) is achieved by M^*, thus the equivalence of the primal and dual problems. $\square$

Remark. It is easy to see that (8) holds even without expectation $\mathbb{E}^{\mathbb{Q}}[\cdot]$ on the left-hand side, i.e.

$$\sup_{t \leq s \leq T} (D_{t,s}F_s - M_s^*) = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}} \left[D_{t,\tau}F_\tau | \mathcal{F}_t \right] \quad \text{almost surely}$$

Lower and Upper bounds

Let u_0 be the price of an American option at time $t = 0$, i.e.,

$$u_0 = \sup_{\tau \in \mathcal{T}_{0,T}} \mathbb{E}^{\mathbb{Q}} \left[D_{0,\tau} F_\tau \right] = \inf_{M \in \mathcal{M}_{0,0}} \mathbb{E}^{\mathbb{Q}} \left[\sup_{0 \le t \le T} \left(D_{0,t} F_t - M_t \right) \right]$$

- Any stopping time (exercise strategy) $\tau \in \mathcal{T}_{0,T}$ gives a lower bound:

$$\mathbb{E}^{\mathbb{Q}} \left[D_{0,\tau} F_\tau \right] \le u_0$$

- Any martingale (self-financing hedge) M with $M_0 = 0$ gives an upper bound:

$$u_0 \le \mathbb{E}^{\mathbb{Q}} \left[\sup_{0 \le t \le T} \left(D_{0,t} F_t - M_t \right) \right].$$

$$\boxed{\text{Primal problem} \Longrightarrow \text{Lower bound}, \quad \text{Dual problem} \Longrightarrow \text{Upper bound}}$$

Upper bound

Suppose an agent sells an American option at time 0 and let M be the value of the self-financing hedge less its initial value, then

$$\mathbb{E}^{\mathbb{Q}}\left[\sup_{0\leq t\leq T}\left(D_{0,t}F_t - M_t\right)\right]$$

gives the maximum exposure from the short option position and its hedge on average.

Example 1. Let $M_t \equiv 0$ (no hedge), i.e.

$$u_0 \leq \mathbb{E}^{\mathbb{Q}}\left[\sup_{0\leq t\leq T}\left(D_{0,t}F_t\right)\right].$$

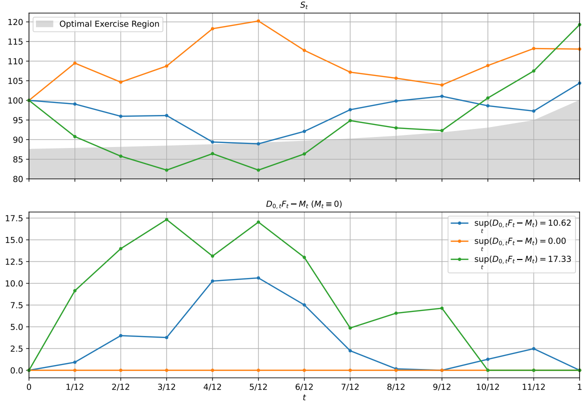
Example 2. In case of American put option, let M_t be the European put price (discounted to time 0) with the same final maturity and strike less the initial price.

Upper Bound - Examples of Different Hedges

Bermudan put, monthly exercises, Black-Scholes model

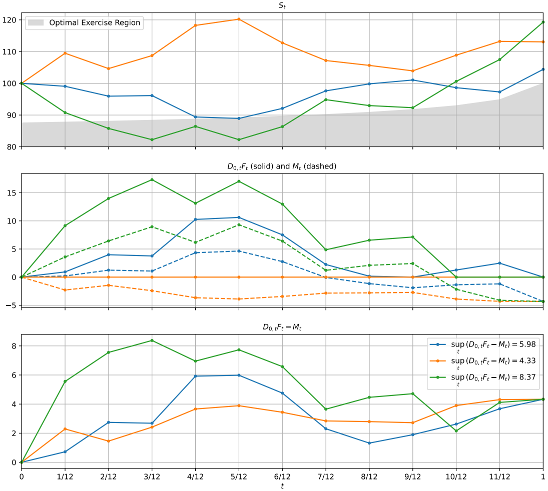
$$S_0 = 100, \sigma = 0.2, r = 0.1, q = 0.02, K = 100$$

- No Hedge:



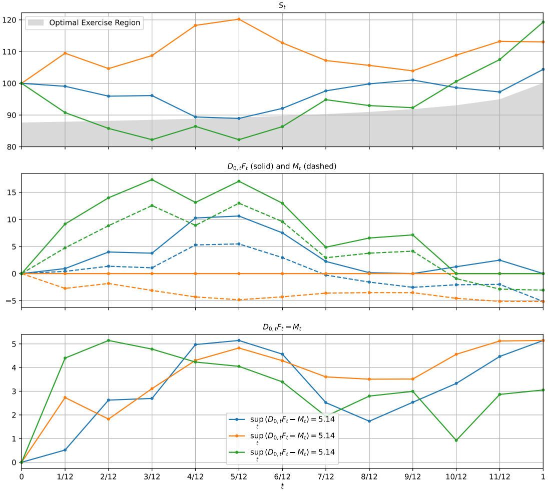
Upper Bound - Examples of Different Hedges

- Hedge with a European Put:



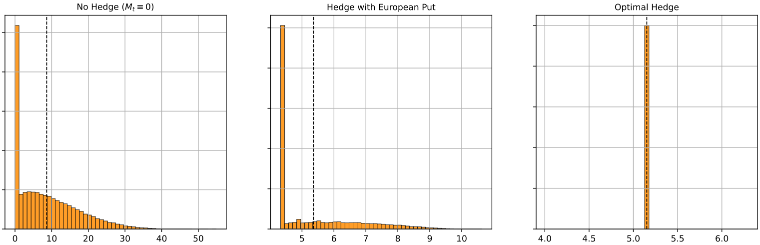
Upper Bound - Examples of Different Hedges

- Optimal Hedge:



Upper Bound - Examples of Different Hedges

- Histogram of $\sup_t (D_{0,t}F_t - M_t)$:



Doob-Meyer Decomposition - Discrete Time

Theorem (Doob Decomposition)

Let U_n be an adapted and integrable process. Then there exists a unique decomposition

$$U_n = M_n - A_n,$$

where M_n is a martingale and A_n is predictable process with $A_0 = 0$. In particular, U_n is a supermartingale if and only if A_n is increasing; U_n is a submartingale if and only if A_n is decreasing.

Proof.

Put $M_0 = U_0$ and $A_0 = 0$. For $n \geq 0$, let

$$M_{n+1} - M_n = U_{n+1} - \mathbb{E}\left[U_{n+1} | \mathcal{F}_n\right]$$
$$A_{n+1} - A_n = -\mathbb{E}\left[U_{n+1} | \mathcal{F}_n\right] + U_n.$$

Then obviously M_n is a martingale and $A_n \in \mathcal{F}_{n-1}$. □

Dual Method: Broadie-Andersen algorithm

Consider a Bermudan option that pays F_{t_i} if exercised at t_i, $i = 1, \ldots, N$.

Step 1

First run a primal algorithm with n_1 paths (e.g. the Longstaff-Schwartz algorithm) to obtain a sequence of (nearly) optimal stopping times τ_i, $i = 1, \ldots, N$. Let V be the (discounted) value process stopped by τ_i, i.e.

$$V_{t_i} = \mathbb{E}^{\mathbb{Q}} \left[D_{0,\tau_i} F_{\tau_i} \middle| \mathcal{F}_{t_i} \right]$$

We shall use V_t as an approximation to the Snell envelope $\mathcal{S}_t$ of $D_{0,t} F_t$ and then extract the martingale part of V_t as an approximation to the optimal martingale M_t^*, which is the martingale part of $\mathcal{S}_t$.

Step 2

Simulate a new set of n_2 independent paths. For each of these paths and at each exercise date t_i, we need to estimate V_{t_i} and $\mathbb{E}^{\mathbb{Q}} \left[V_{t_{i+1}} | \mathcal{F}_{t_i} \right]$.

Dual Method: Broadie-Andersen algorithm

- If τ_i indicates continuation, i.e. $\tau_i > t_i$, then simulate n_c independent subpaths starting from (t_i, X_{t_i}) and stopped by τ_i

$$V_{t_i} = \mathbb{E}^{\mathbb{Q}}\left[V_{t_{i+1}} | \mathcal{F}_{t_i}\right] \approx \frac{1}{n_c} \sum_{j=1}^{n_c} D_{0,\tau_i}^{(j)} F_{\tau_i}^{(j)}$$

- If τ_i indicates exercise, i.e. $\tau_i = t_i$, then $V_{t_i} = D_{0,t_i} F_{t_i}$, but we still need to estimate $\mathbb{E}^{\mathbb{Q}}\left[V_{t_{i+1}} | \mathcal{F}_{t_i}\right]$. Simulate n_e independent subpaths starting from (t_i, X_{t_i}) and stopped at τ_{i+1},

$$\mathbb{E}^{\mathbb{Q}}\left[V_{t_{i+1}} \middle| \mathcal{F}_{t_i}\right] \approx \frac{1}{n_e} \sum_{j=1}^{n_e} D_{0,\tau_{i+1}}^{(j)} F_{\tau_{i+1}}^{(j)}$$

Step 3

Build the martingale M: $M_0 = 0$ and

$$M_{t_{i+1}} = M_{t_i} + V_{t_{i+1}} - \mathbb{E}^{\mathbb{Q}}\left[V_{t_{i+1}} | \mathcal{F}_{t_i}\right]$$

and then obtain the upper bound price

$$\mathbb{E}^{\mathbb{Q}}\left[\max_{1 \le i \le N} \left(D_{0,t_i} F_{t_i} - M_{t_i}\right)\right]$$

Dual method: Broadie-Andersen algorithm

- The nested simulations in Step 2 are not recursive, i.e., no further simulations are run on each subpath. The number of simulated paths is at most
$$n_1 + n_2 \times (N-1) \times \max(n_c, n_e)$$
- For the martingale (M_{t_i}) generated from a nearly optimal strategy, $\max_{1\leq i\leq N}(D_{0,t_i}F_{t_i} - M_{t_i})$ has very small variance. Therefore, in most cases, even small values of n_2 give accurate results.
- A large duality gap is usually an indication of poor exercise rules from the primal algorithm (often caused by regression problems, such as bad selection of basis functions, over-fitting, etc.)
- The nested Monte Carlo generates independent sampling error with zero mean. However, applying the $\sup$ operator subsequently produces a high-biased estimator of the upper bound.

Option Pricing in a Nutshell

Option Pricing in a Nutshell

In this chapter:

- We recall classical results on stochastic representations of solutions of linear parabolic PDEs.
- This allows us to revisit key notions of option pricing and set our notations.
- Most of the material presented here is standard and can be found in various classical references. Yet we also highlight several important notions which are not often covered:
  - super-replication
  - pricing in incomplete models

1. The superreplication paradigm

Models of financial markets

- A filtered probability space $\left(\Omega, (\mathcal{F}_t)_{0\leq t\leq T}, \mathbb{P}^{\mathrm{hist}}\right)$
- $\mathbb{P}^{\mathrm{hist}}$ is the historical or real probability measure under which we model our market.
- A market model is defined by an n-dimensional stochastic differential equation (SDE)

$$dX_t^i = b_i(t, X_t)\,dt + \sum_{j=1}^{d} \sigma_{i,j}(t, X_t)\,dW_t^j, \qquad i \in \{1, \ldots, n\} \tag{9}$$

and by another positive stochastic process B_t, called the money-market account, representing the value of cash, which satisfies

$$dB_t = r_t B_t\,dt, \qquad B_0 = 1$$

i.e.,

$$B_t = \exp\left(\int_0^t r_s\,ds\right)$$

- r_t is the short term interest rate. It is adapted to $\mathcal{F}_t$, which is the (natural) filtration generated by the d-dimensional uncorrelated standard Brownian motion $\{W_t^j\}_{1\leq j\leq d}$.

Models of financial markets

- In order to ensure that SDE (9) admits a unique strong solution (see e.g., Karatzas), we assume that b and σ satisfy:

  Assum(SDE): The functions b and σ are Lipschitz-continuous in x uniformly in t, and satisfy a linear growth condition: there exists a positive constant C such that for all $t \geq 0$, $x, y \in \mathbb{R}^n$,

$$\begin{aligned} |b(t,x) - b(t,y)| + |\sigma(t,x) - \sigma(t,y)| &\leq C|x-y| \\ |b(t,x)| + |\sigma(t,x)| &\leq C\left(1 + |x|\right) \end{aligned}$$

- We set

$$D_{tu} := B_t B_u^{-1} = \exp\left(-\int_t^u r_s \, ds\right)$$

  which is the discount factor from date u to date t.
- Throughout the course, we will denote by $\tilde{Y}_t := D_{0t} Y_t$ the discounted value of any price process Y_t.
- Certain market components X^i may not be sold or bought in the market, such as the short term interest rate, or a stochastic volatility. Throughout this course, a market component X^i that can be sold and bought in the market is called an "asset."

Self-financing portfolios

- Let us assume that we have a portfolio consisting of m assets, say $X_t^1, \ldots, X_t^m$, and the money-market account B_t. It is convenient to use the notation X^0 for B.
- The portfolio at a time t is composed of Δ_t^i assets X_t^i and Δ_t^0 units of X_t^0 (cash).
- The Δ_t^i's must be $\mathcal{F}_t$-measurable, i.e., we cannot look into the future.
- The portfolio value π_t is

$$\pi_t := \sum_{i=0}^{m} \Delta_t^i X_t^i \tag{10}$$

- As time passes, we can readjust the allocations Δ_t^i, but no cash is ever injected into or removed from the portfolio: between t and $t + dt$, the variation in the portfolio value is only due to the variation of the values of the assets, i.e.,

$$d\pi_t = \sum_{i=0}^{m} \Delta_t^i \, dX_t^i \tag{11}$$

We then speak of a *self-financing portfolio*.

Self-financing portfolios

- In terms of discounted values, this reads

$$d\tilde{\pi}_t = \sum_{i=0}^{m} \Delta_t^i \, d\tilde{X}_t^i = \sum_{i=1}^{m} \Delta_t^i \, d\tilde{X}_t^i \tag{12}$$

because for any price process Y_t, $d\tilde{Y}_t = D_{0t}(dY_t - r_t Y_t \, dt)$, so

$$\tilde{\pi}_t = \pi_0 + \sum_{i=1}^{m} \int_0^t \Delta_s^i \, d\tilde{X}_s^i = \pi_0 + \int_0^t \Delta_s \cdot d\tilde{X}_s \tag{13}$$

where $\cdot$ denotes the usual scalar product in $\mathbb{R}^m$.[1]

- As a technical condition, we need to introduce:

Definition

$(\Delta_t, 0 \le t \le T)$ defines an **admissible** portfolio if $\tilde{\pi}_t$ is bounded from below for all t $\mathbb{P}^{\text{hist}}$-a.s., i.e., there exists $M \in \mathbb{R}$ such that

$$\mathbb{P}^{\text{hist}} \left(\forall t \in [0, T], \ \tilde{\pi}_t \ge M \right) = 1$$

[1] The stochastic integral is well-defined with the condition $\int_0^t (\Delta_s^i)^2 \, d\langle X \rangle_s < \infty$ $\mathbb{P}^{\text{hist}}$-a.s. $\tilde{\pi}_t$ is then a local martingale.

Arbitrage and arbitrage-free models

- An arbitrage is a self-financing strategy that is worth zero initially and yields a positive gain without any risk:

Definition

A self-financing admissible portfolio is called an **arbitrage** if the corresponding value process π_t satisfies $\pi_0 = 0$ and

$$\pi_T \geq 0 \quad \mathbb{P}^{\mathrm{hist}} - \mathrm{a.s} \qquad \text{and} \qquad \mathbb{P}^{\mathrm{hist}}(\pi_T > 0) > 0$$

- Arbitrageurs are a special kind of traders. Their role is precisely to detect and take full advantage of arbitrage opportunities as soon as they appear in the market. This impacts market prices: arbitrage opportunities tend to disappear as soon as they arise. Absence of arbitrage opportunities is therefore a natural modeling assumption.

Arbitrage and arbitrage-free models

- The next lemma gives a sufficient condition under which we exclude arbitrage opportunities in our market model:

Lemma (Sufficient condition excluding arbitrage)

Suppose there exists a measure $\mathbb{Q}$ on $(\Omega, \mathcal{F}_T)$ such that $\mathbb{Q} \sim \mathbb{P}^{\text{hist}}$ and such that, for all **asset** X^i*, the discounted price process $\{\tilde{X}^i_t\}_{t\in[0,T]}$ is a local martingale with respect to $\mathbb{Q}$. Then the market $\{X_t\}_{t\in[0,T]}$ has no arbitrage.*

- $\mathbb{P}$ is equivalent to $\mathbb{Q}$ (denoted by $\mathbb{P} \sim \mathbb{Q}$) if and only if $\forall A \in \mathcal{F}_T$, $\mathbb{P}(A) = 0 \Longleftrightarrow \mathbb{Q}(A) = 0$.
- Throughout this course, unless stated otherwise, martingales and local martingales are considered with respect to the filtration $(\mathcal{F}_t)_{0\le t\le T}$.
- Note that the assumption of this lemma bears only on *assets* X^i only, not on non-tradable components of X, such as instantaneous interest rates, instantaneous stochastic volatility, etc.

Arbitrage and arbitrage-free models

Proof.

Let us suppose that there exists an arbitrage. We have

$$\tilde{\pi}_t = \pi_0 + \sum_{i=1}^{m} \int_0^t \Delta_s^i \, d\tilde{X}_s^i = \pi_0 + \int_0^t \Delta_s \cdot d\tilde{X}_s$$

and, for all asset X^i, the discounted price process $\{\tilde{X}_t^i\}_{t\in[0,T]}$ is a local martingale with respect to $\mathbb{Q}$. Therefore $\tilde{\pi}_t$ is a local martingale w.r.t. $\mathbb{Q}$. Moreover, Δ defines an admissible portfolio, so $\tilde{\pi}_t$ is a lower bounded local martingale w.r.t. $\mathbb{Q}$. By a standard result, $\tilde{\pi}_t$ is thus a $\mathbb{Q}$-supermartingale and hence $\mathbb{E}^{\mathbb{Q}}[\tilde{\pi}_T] \leq \tilde{\pi}_0 = 0$. But since $\tilde{\pi}_T \geq 0$ $\mathbb{P}^{\mathrm{hist}}$-a.s. and $\mathbb{Q} \sim \mathbb{P}^{\mathrm{hist}}$, we have $\tilde{\pi}_T \geq 0$ $\mathbb{Q}$-a.s. Hence $\tilde{\pi}_T = 0$ $\mathbb{Q}$-a.s., so $\tilde{\pi}_T = 0$ $\mathbb{P}^{\mathrm{hist}}$-a.s., which yields $\pi_T = 0$ $\mathbb{P}^{\mathrm{hist}}$-a.s. This contradicts the fact that $\mathbb{P}^{\mathrm{hist}}(\pi_T > 0) > 0$. □

Arbitrage and arbitrage-free models

- This simple lemma invites one to introduce the notion of an equivalent local martingale measure (in short ELMM):

Definition (Equivalent local martingale measure)

Any measure $\mathbb{Q} \sim \mathbb{P}^{\text{hist}}$ on $(\Omega, \mathcal{F}_T)$ such that, for all *asset* X^i, the discounted price process $\{\tilde{X}^i_t\}_{t\in[0,T]}$ is a local martingale w.r.t. $\mathbb{Q}$ is called an equivalent local martingale measure (ELMM). An ELMM is also called a risk-neutral measure.

- The previous lemma now reads: if there exists an ELMM, then the market has no arbitrage opportunities.
- Throughout this course we will always assume that we are in a situation where there exists at least one ELMM.
- Let us consider an *asset* X^i. Under an ELMM $\mathbb{Q}$, $\{\tilde{X}^i_t\}_{t\in[0,T]}$ is an $(\mathcal{F}_t)$-local martingale, hence has zero drift. As a consequence, the drift of the *asset* X^i under an ELMM is $r_t X^i_t$:

$$dX^i_t = \exp\left(\int_0^t r_s\, ds\right)(d\tilde{X}^i_t + r_t \tilde{X}^i_t\, dt) = r_t X^i_t\, dt + (\cdots)\cdot dW_t$$

This is not the case for non-tradable components of the market.

Super-replication

- Let us assume that, at time t, we buy and delta-hedge a European option written on m assets, say $X_t^1, \ldots, X_t^m$, with maturity T and payoff F_T, at the price z.
- In general, the payoff F_T is a function of the paths $(X_t^i, 0 \leq t \leq T)$ followed by the prices of the m assets between times 0 and T.
- The final value of the buyer's portfolio, discounted at time 0, is

$$\begin{aligned}\tilde{\pi}_T^B &= -D_{0t} z + \sum_{i=1}^m \int_t^T \Delta_s^i \, d\tilde{X}_s^i + D_{0T} F_T \\ &= -D_{0t} z + \int_t^T \Delta_s \cdot d\tilde{X}_s + D_{0T} F_T\end{aligned}$$

We then define the buyer's super-replication price at time t as the greatest price z s.t. the value of the buyer's portfolio $\tilde{\pi}_T^B$ is $\mathbb{P}^{\mathrm{hist}}$-a.s. nonnegative:

Super-replication

Definition (Buyer's price)

$$\mathcal{B}_t(F_T) = \sup \Big\{ z \in \mathcal{F}_t \;\Big|\; \text{there exists an admissible portfolio } \Delta \text{ such that}$$
$$\tilde{\pi}^B_T := -D_{0t} z + \int_t^T \Delta_s \cdot d\tilde{X}_s + D_{0T} F_T \geq 0 \;\; \mathbb{P}^{\text{hist}} - a.s. \Big\} \qquad (14)$$

- The price z must be $\mathcal{F}_t$-measurable, denoted by $z \in \mathcal{F}_t$, i.e., we cannot look into the future.
- Similarly, we can define the seller's super-replication price as:

Definition (Seller's price)

$$\mathcal{S}_t(F_T) = \inf \Big\{ z \in \mathcal{F}_t \;\Big|\; \text{there exists an admissible portfolio } \Delta \text{ such that}$$
$$\tilde{\pi}^S_T := D_{0t} z + \int_t^T \Delta_s \cdot d\tilde{X}_s - D_{0T} F_T \geq 0 \;\; \mathbb{P}^{\text{hist}} - a.s. \Big\} \qquad (15)$$

Super-replication

Theorem (Arbitrage-free bounds)

Assume that there exists an ELMM $\mathbb{Q} \sim \mathbb{P}^{\text{hist}}$. *Then*

$$\mathcal{B}_t(F_T) \leq \mathbb{E}^{\mathbb{Q}}[D_{tT} F_T | \mathcal{F}_t] \leq \mathcal{S}_t(F_T)$$

Proof.

From the definition of $\mathcal{B}_t(F_T)$, we assume that there exists an admissible portfolio Δ such that

$$-D_{0t} z + \int_t^T \Delta_s \cdot d\tilde{X}_s + D_{0T} F_T \geq 0 \qquad \mathbb{P}^{\text{hist}} - a.s.$$

$\int_0^t \Delta_s \cdot d\tilde{X}_s$ is a lower bounded $\mathbb{Q}$-local martingale, hence a $\mathbb{Q}$-supermartingale, so $\mathbb{E}^{\mathbb{Q}}[\int_t^T \Delta_s \cdot d\tilde{X}_s | \mathcal{F}_t] \leq 0$ and we deduce that

$$D_{0t} z \leq \mathbb{E}^{\mathbb{Q}}[D_{0T} F_T | \mathcal{F}_t] \qquad \text{i.e.,} \qquad z \leq \mathbb{E}^{\mathbb{Q}}[D_{tT} F_T | \mathcal{F}_t]$$

Taking the supremum over z, we get $\mathcal{B}_t(F_T) \leq \mathbb{E}^{\mathbb{Q}}[D_{tT} F_T | \mathcal{F}_t]$. The inequality $\mathbb{E}^{\mathbb{Q}}[D_{tT} F_T | \mathcal{F}_t] \leq \mathcal{S}_t(F_T)$ can be derived similarly. □

Attainable payoff

- These bounds can be sharpened by assuming that the market is complete.
- In order to define what it means for a market to be complete, we need to introduce the notion of attainable payoff:

Definition (Attainable payoff)

A payoff F_T is said to be attainable (at time 0) if there exists an admissible portfolio Δ and a real number z such that $z + \int_0^T \Delta_s \cdot d\tilde{X}_s - D_{0T} F_T = 0$ $\mathbb{P}^{\text{hist}}$-a.s. and $\int_0^t \Delta_s \cdot d\tilde{X}_s$ is a (true) $\mathbb{Q}$-martingale with $\mathbb{Q} \sim \mathbb{P}^{\text{hist}}$.

- Stated otherwise, a payoff F_T is said to be attainable if we can generate the wealth F_T at time T from an initial wealth z by trading only in the underlying assets and the cash in a self-financing way.
- The portfolio Δ and the real number z are unique because if $z' + \int_0^T \Delta'_s \cdot d\tilde{X}_s - D_{0T} F_T = 0$, then

$$\int_0^T (\Delta'_s - \Delta_s) \cdot d\tilde{X}_s = z - z'$$

and since $\tilde{X}$ is a (nontrivial) $\mathbb{Q}$-local martingale, this yields $\Delta'_s = \Delta_s$, hence $z' = z$.

Complete markets

- As a consequence, there is a unique fair price at $t = 0$ for an attainable payoff, and this price is z.
- The assumption that $\int_0^t \Delta_s \cdot d\tilde{X}_s$ is a (true) $\mathbb{Q}$-martingale guarantees that $z = \mathbb{E}^{\mathbb{Q}}[D_{0T} F_T]$, so we have a formula to compute the price z from the payoff F_T.

Definition (Complete market)

A market is said to be complete when every payoff is attainable at time 0.

Complete markets

Theorem

For a complete market, for any ELMM $\mathbb{Q}$ and any payoff F_T, we have

$$\mathcal{B}_t(F_T) = \mathcal{S}_t(F_T) = \mathbb{E}^{\mathbb{Q}}[D_{tT} F_T | \mathcal{F}_t]$$

Proof.

By completeness, we can find a real number z and an admissible portfolio Δ such that $z + \int_0^T \Delta_s \cdot d\tilde{X}_s - D_{0T} F_T = 0$ $\mathbb{P}^{\text{hist}}$-a.s., hence $\mathbb{Q}$-a.s. As a consequence,

$$D_{0t} z_t + \int_t^T \Delta_s \cdot d\tilde{X}_s - D_{0T} F_T = 0 \qquad \mathbb{Q}\text{−a.s.}$$

with $z_t = D_{0t}^{-1} \left(z + \int_0^t \Delta_s \cdot d\tilde{X}_s \right)$ being $\mathcal{F}_t$-measurable. Since $\int_0^t \Delta_s \cdot d\tilde{X}_s$ is a $\mathbb{Q}$-martingale, we have $D_{0t} z_t = \mathbb{E}^{\mathbb{Q}}[D_{0T} F_T | \mathcal{F}_t]$, i.e., $z_t = \mathbb{E}^{\mathbb{Q}}[D_{tT} F_T | \mathcal{F}_t]$, whence $\mathcal{B}_t(F_T) \geq \mathbb{E}^{\mathbb{Q}}[D_{tT} F_T | \mathcal{F}_t]$. Combined with the reverse inequality, we get $\mathcal{B}_t(F_T) = \mathbb{E}^{\mathbb{Q}}[D_{tT} F_T | \mathcal{F}_t]$. We proceed similarly for the other equality. $\square$

Super-replication in incomplete markets

- With incomplete markets or markets with friction, perfect replication is no longer possible.
- In the theorem below, we state that the buyer's super-replication price is given by the infimum of expectations of the discounted payoff over the set ELMM of equivalent local martingale measures:

Theorem (Super-replication price (El Karoui-Quenez, Kramkov))

The buyer's super-replication price and the seller's super-replication price are given by

$$\mathcal{B}_t(F_T) = \inf_{\mathbb{Q}\in\mathrm{ELMM}} \mathbb{E}^{\mathbb{Q}}[D_{tT}F_T|\mathcal{F}_t]$$

$$\mathcal{S}_t(F_T) = \sup_{\mathbb{Q}\in\mathrm{ELMM}} \mathbb{E}^{\mathbb{Q}}[D_{tT}F_T|\mathcal{F}_t]$$

Characterization of complete markets

Corollary (Complete market)

A market is complete if and only if there exists a unique ELMM.

Proof.

Assume the market is complete. Let $\mathbb{Q}^0$ be an ELMM. For any payoff F_T,

$$\mathcal{B}_0(F_T) = \mathcal{S}_0(F_T) = \mathbb{E}^{\mathbb{Q}^0}[D_{0T}F_T] = \inf_{\mathbb{Q}\in\mathrm{ELMM}} \mathbb{E}^{\mathbb{Q}}[D_{0T}F_T] = \sup_{\mathbb{Q}\in\mathrm{ELMM}} \mathbb{E}^{\mathbb{Q}}[D_{0T}F_T].$$

Hence, for all F_T, $\mathbb{E}^{\mathbb{Q}}[D_{0T}F_T] = \mathbb{E}^{\mathbb{Q}^0}[D_{0T}F_T]$ for all $\mathbb{Q} \in \mathrm{ELMM}$, so that $\mathrm{ELMM} = \{\mathbb{Q}^0\}$. Conversely, if $\mathrm{ELMM} = \{\mathbb{Q}^0\}$, then $\mathcal{B}_0(F_T) = \mathcal{S}_0(F_T) = \mathbb{E}^{\mathbb{Q}^0}[D_{0T}F_T]$. As a consequence, there exist two admissible portfolios Δ^B and Δ^S (one for the buyer and one for the seller) such that $\mathbb{P}^{\mathrm{hist}}$-a.s., i.e., $\mathbb{Q}^0$-a.s.

$$\begin{aligned} -\mathbb{E}^{\mathbb{Q}^0}[D_{0T}F_T] + \int_0^T \Delta_t^B \cdot d\tilde{X}_t + D_{0T}F_T &\geq 0 \\ \mathbb{E}^{\mathbb{Q}^0}[D_{0T}F_T] + \int_0^T \Delta_t^S \cdot d\tilde{X}_t - D_{0T}F_T &\geq 0 \end{aligned}$$

Summing those inequalities, this means that $\mathbb{Q}^0$-a.s. $\int_0^T (\Delta_t^B + \Delta_t^S) \cdot d\tilde{X}_t \geq 0$, and, because the discounted prices of all *assets* $(\tilde{X}_t^i)$ are $\mathbb{Q}^0$-local martingales, this yields $\Delta_t^B = -\Delta_t^S$ $\mathbb{Q}^0$-a.s. Eventually, we have that $\mathbb{E}^{\mathbb{Q}^0}[D_{0T}F_T] + \int_0^T \Delta_t^S \cdot d\tilde{X}_t - D_{0T}F_T = 0$ $\mathbb{Q}^0$-a.s., hence $\mathbb{P}^{\mathrm{hist}}$-a.s., and the market is complete. □

Complete models versus incomplete models

- From the proof of the super-replication price theorem, the market model (9) is complete if and only if there exists a unique $\mathcal{F}_t$-adapted vector $\lambda_t \in \mathbb{R}^d$ s.t.

$$\forall i \in \text{asset}, \qquad b_i(t, X_t) - r_t X_t^i = \sum_{j=1}^{d} \sigma_{i,j}(t, X_t)\lambda_t^j \tag{16}$$

- The unique ELMM $\mathbb{Q}$ is then given by

$$\frac{d\mathbb{Q}}{d\mathbb{P}^{\text{hist}}}_{|\mathcal{F}_T} = \prod_{j=1}^{d} e^{-\int_0^T \lambda_t^j \, dW_t^j - \frac{1}{2}\int_0^T \left(\lambda_t^j\right)^2 dt}$$

- The unique arbitrage-free price is

$$\mathcal{B}_t(F_T) = \mathcal{S}_t(F_T) = \mathbb{E}^{\mathbb{Q}}[D_{tT} F_T | \mathcal{F}_t]$$

- An inspection of (16) reveals that if the market is complete, then the rank of $\sigma(t, X_t)$ is equal to d a.s. (which implies that $\#\text{assets} \geq d$). In the case where $\#\text{assets} < d$, the market cannot be complete.
- Moreover, if the number of assets coincides with the number of Brownian motions, i.e., $\#\text{assets} = d$, and $(\sigma_{i,j}(t, X_t))_{i \in \text{asset}, 1 \leq j \leq d}$ is invertible, then the market is complete.

Complete models versus incomplete models

- Examples of complete models that are commonly used by practitioners include
  - Dupire's local volatility model (see Dupire)
  - Libor market models with local volatilities, e.g., BGM with deterministic volatilities (see Brace-Gatarek-Musiela)
  - and Markov functional models (see Hunt-Kennedy-Pelsser).
- Common examples of incomplete models are stochastic volatility models (in short SVMs). Here $\#\text{assets} < d$.
- An example of stochastic volatility model is the Heston model. The dynamics of the underlying, denoted by X_t, reads under a risk-neutral measure $\mathbb{Q}^0 \sim \mathbb{P}^{\text{hist}}$ as

$$\frac{dX_t}{X_t} = r_t\,dt + \sqrt{V_t}\left(\rho\,dW_t^2 + \sqrt{1-\rho^2}\,dW_t^1\right) \tag{17}$$

$$dV_t = -k(V_t - m)\,dt + \sigma\sqrt{V_t}\,dW_t^2 \tag{18}$$

with W_t^1, W_t^2, two uncorrelated standard $\mathbb{Q}^0$-Brownian motions.

Complete models versus incomplete models

- V_t is the instantaneous variance, it is *not* a tradable instrument. The only asset is X_t. Its drift under any ELMM is $r_t X_t$.
- The incompleteness of such a model can be detected as the drift term in V_t can be arbitrarily modified with a change of measure from $\mathbb{Q}^0$ to $\mathbb{Q}^1 \sim \mathbb{P}^{\text{hist}}$:

$$\frac{d\mathbb{Q}^1}{d\mathbb{Q}^0}_{|\mathcal{F}_T} = \prod_{j=1}^{2} e^{-\int_0^T \lambda_t^j \, dW_t^j - \frac{1}{2}\int_0^T \left(\lambda_t^j\right)^2 dt}$$

such that

$$\rho \lambda_t^2 + \sqrt{1-\rho^2}\lambda_t^1 = 0 \tag{19}$$

- From Girsanov's theorem, Equation (19) guarantees that the drift of X_t under $\mathbb{Q}^1$ is still $r_t X_t$, i.e., that $\mathbb{Q}^1$ is still an ELMM.

Complete models versus incomplete models

- We can show that the seller's super-replication (undiscounted) price at $t=0$ of a European vanilla payoff $F_T := g(X_T)$ in such an SVM is $g^{\mathrm{conc}}(X_0)$ with g^{conc} the concave envelope of g, i.e., the smallest concave function that is greater than or equal to g.
- Similarly, the buyer's super-replication (undiscounted) price at $t=0$ of a European vanilla payoff $g(X_T)$ in such an SVM is $g^{\mathrm{conv}}(X_0)$ with g^{conv} the convex envelope of g, i.e., the largest convex function that is smaller than or equal to g.
- For example, for a call option $g(x)=(x-K)^+$, $g^{\mathrm{conc}}(x)=x$ and $g^{\mathrm{conv}}(x)=(x-K)^+$.
- This simple example illustrates the main drawback of the super-replication framework: the buyer's and seller's super-replication prices are not close and therefore cannot be used in practice.[2]

[2] This drawback can be circumvented by adding extra market instruments, such as vanilla options, in the superhedging strategy. This leads to tight bounds and a nice generalization of the optimal transport theory (see Beiglböck-Henry-Labordère-Penkner, Galichon-Henry-Labordère-Touzi) where now the transport is performed along a martingale measure.

Pricing in practice

- In practice, the seller's price at time t is computed by picking out a particular ELMM $\mathbb{Q}$:

$$u_t := \mathbb{E}^{\mathbb{Q}}[D_{tT} F_T | \mathcal{F}_t] \tag{20}$$

  Under $\mathbb{Q}$, the drift for an asset X^i is fixed to $b_i(t, X_t) = r_t X_t^i$.
- In an incomplete market, $\mathbb{Q}$ does not necessarily achieve the supremum in the super-replication price theorem, and we lose the superhedging strategy paradigm. Selling options becomes a risky business.
- In practice, picking a particular ELMM simplifies a lot the pricing problem: it becomes a *linear* problem: the price of the (European) payoff $F_T^1 + F_T^2$ equals the sum of the prices of the (European) payoffs F_T^1 and F_T^2.

2. Stochastic representation of solutions of linear PDEs

The Cauchy problem

- We will always assume that there exists (deterministic) function r such that $r_t = r(t, X_t)$. $D_{t_1 t_2} = \exp\left(-\int_{t_1}^{t_2} r(s, X_s)\, ds\right)$
- We denote by $\mathcal{L}$ the Itô generator of X defined under a risk-neutral measure $\mathbb{Q}$:

$$\mathcal{L} = \sum_{i=1}^{n} b_i(t,x)\partial_i + \frac{1}{2}\sum_{i,j=1}^{n}\sum_{k=1}^{d} \sigma_{i,k}(t,x)\sigma_{j,k}(t,x)\partial_{ij} \tag{21}$$

If X^i is an asset, $b_i(t,x) = r(t,x)x_i$.

- If there exists g such that $F_T = g(X_T)$, we speak of a vanilla option. In such a case, by the Markov property of X,

$$u_t = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r(s, X_s)\, ds\right) g(X_T) \middle| \mathcal{F}_t\right] =: u(t, X_t)$$

is a function u of (t, X_t); the discounted price process

$$D_{0t}u_t = D_{0t}\mathbb{E}^{\mathbb{Q}}[D_{tT}F_T|\mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}}[D_{0T}F_T|\mathcal{F}_t]$$

is a $\mathbb{Q}$-martingale. Then a straightforward application of Itô's lemma to $D_{0t}u(t, X_t) = \exp\left(-\int_0^t r(s, X_s)\, ds\right) u(t, X_t)$ shows that the pricing function u, if smooth enough, satisfies the second order parabolic linear PDE:

$$\partial_t u(t,x) + \mathcal{L}u(t,x) - r(t,x)u(t,x) = 0 \tag{22}$$

with the terminal condition $u(T,x) = g(x)$.

The Cauchy problem

- Consider the case where there exists f, g such that the payoffs are $g(X_T)$ at maturity T and $f(t, X_t)\,dt$ between t and $t + dt$ ($0 \le t \le T$).
- f: stream of continuous cash flows; think of daily, weekly, or monthly coupons.
- By the Markov property of X,

$$u_t = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r(s, X_s)\,ds\right) g(X_T) + \int_t^T \exp\left(-\int_t^s r(a, X_a)\,da\right) f(s, X_s)\,ds \middle| \mathcal{F}_t\right]$$

is a function u of (t, X_t); the discounted price process

$$\begin{aligned} D_{0t}u_t + \int_0^t D_{0s} f(s, X_s)\,ds &= D_{0t}\mathbb{E}^{\mathbb{Q}}\left[D_{tT}F_T + \int_0^T D_{0s} f(s, X_s)\,ds \middle| \mathcal{F}_t\right] \\ &= \mathbb{E}^{\mathbb{Q}}\left[D_{0T}F_T + \int_0^T D_{0s} f(s, X_s)\,ds \middle| \mathcal{F}_t\right] \end{aligned}$$

is a $\mathbb{Q}$-martingale. Then a straightforward application of Itô's lemma to $D_{0t}u(t, X_t) + \int_0^t D_{0s} f(s, X_s)\,ds$ shows that the pricing function u, if smooth enough, satisfies the second order parabolic linear PDE:

$$\partial_t u(t, x) + \mathcal{L}u(t, x) - r(t, x)u(t, x) + f(t, x) = 0 \tag{23}$$

with the terminal condition $u(T, x) = g(x)$.

- $f(t, x)$ is called "source term."
- The case with dividend/repo yield $q(t, X_t)$.

The Cauchy problem

Conversely, a solution u to (23) admits the stochastic representation (20) as stated by the Feynman-Kac theorem:

Theorem (Feynman-Kac, see e.g., Karatzas)

Let b, σ satisfy **Assum(SDE)**, *r be uniformly bounded from below and f have quadratic growth in x uniformly in t. Let $u \in C^{1,2}([0,T) \times \mathbb{R}^d) \cap C^0([0,T] \times \mathbb{R}^d)$ with quadratic growth in x uniformly in t and solution to the parabolic PDE:*

$$\partial_t u(t,x) + \mathcal{L}u(t,x) - r(t,x)u(t,x) + f(t,x) = 0$$

with terminal condition $u(T,x) = g(x)$. Then u admits the stochastic representation

$$u(t,x) = \mathbb{E}^{\mathbb{Q}}\left[\int_t^T e^{-\int_t^s r(u,X_u)du} f(s,X_s)ds + e^{-\int_t^T r(s,X_s)ds} g(X_T) \middle| X_t = x\right]$$

Think of f as intermediate payoffs.

Path-dependent options

- Path-dependent payoffs are those payoffs for which one cannot write $F_T = g(X_T)$, i.e., whose value depends on the asset values not only at maturity T, but also at previous dates $t < T$.
- For instance, with one asset $X_t = X_t^1$:
  - Asian options: $F_T = g\left(\frac{1}{T}\int_0^T X_t \, dt\right)$
  - Barrier and lookback options: $F_T = g\left(\min_{t\in[0,T]} X_t, \max_{t\in[0,T]} X_t, X_T\right)$
  - Cliquet options: $F_T = g\left(X_{T_i}/X_{T_{i-1}}, 1 \le i \le N\right)$
- In some cases, one can write SDEs for the path-dependent variables and add them to the SDEs describing the market X so that one can write $F_T = g(X'_T)$ for an enlarged market X', and the option price satisfies a PDE in the variables x'. For instance, the price of an Asian option satisfies a PDE in the variables (x, a) where $A_t = \int_0^t X_s \, ds$ and $dA_t = X_t \, dt$.

Path-dependent options

- For a general path-dependent payoff $F_T = g(X_{t_1}, \ldots, X_{t_n})$, depending on the spot values at the observation dates $t_1 < \cdots < t_n = T$, the PDE, depending on the past spot values $Z^1, \ldots, Z^n$, can be written as

$$\partial_t u_i(t, x, Z^1, \ldots, Z^{i-1}) + \mathcal{L} u_i(t, x, Z^1, \ldots, Z^{i-1}) = 0, \qquad \forall t \in [t_{i-1}, t_i)$$

with the matching conditions at the dates $t_1, \ldots, t_n$,

$$\begin{aligned} u_n(t_n, x, Z^1, \ldots, Z^{n-1}) &= g(Z^1, \ldots, Z^{n-1}, x) \\ u_i(t_i^-, x, Z^1, \ldots, Z^{i-1}) &= u_{i+1}(t_i^+, x, Z^1, \ldots, Z^{i-1}, x) \end{aligned}$$

- The final price is $u_1(0, X_0)$.

Path-dependent options - An example

Example. Consider a path-dependent payoff at time t_n written as

$$g\left(\frac{X_{t_1} + \cdots + X_{t_n}}{n}\right)$$

Introduce an auxiliary variable I such that the pricing function $u(t, x, I)$ satisfies

$$\partial_t u_i(t, x, I) + \mathcal{L}u(t, x, I) = 0, \qquad t_{i-1} < t < t_i, \quad i = 2, \ldots, n$$

with terminal condition at t_n,

$$u(t_n, x, I) = g\left(I + \frac{x}{n}\right)$$

and jump condition at t_i, $i = 1, \ldots, n-1$,

$$u(t_i^-, x, I) = u\left(t_i^+, x, I + \frac{x}{n}\right).$$

Discretize I-values into m different values, and solve m different one-dimensional PDEs backward in time and exchange information with each other at each t_i by the jump condition. Note that in the first time interval $0 < t < t_1$, the auxillary variable I is not needed and the pricing function u simply satisfies

$$\partial_t u(t, x) + \mathcal{L}u(t, x) = 0, \qquad 0 < t < t_1.$$

Finally the price at time 0 is $u(0, X_0)$.

Derivatives with Default Risk

We model the time of default, τ, to be the first jump time of a Poisson process, with deterministic intensity λ_t. That is, for any $t \geq 0$, we have

$$\mathbb{P}\left[\tau \in (t, t+dt) \middle| \tau > t\right] = \lambda_t dt \tag{24}$$

Let $\Phi(t) = \mathbb{P}\left[\tau \leq t\right]$ be the CDF of τ. Then it follows from (24) that

$$\frac{\Phi'(t)dt}{1-\Phi(t)} = \lambda_t dt$$

Solving this ODE gives us the following:

$$\mathbb{P}\left[\tau \leq t\right] = 1 - e^{-\int_0^t \lambda_s ds}$$

$$\mathbb{P}\left[\tau > t\right] = e^{-\int_0^t \lambda_s ds}$$

$$\mathbb{P}\left[\tau \in (t, t+dt)\right] = \lambda_t e^{-\int_0^t \lambda_s ds} dt$$

$$\mathbb{P}\left[\tau > T \middle| \tau > t\right] = e^{-\int_t^T \lambda_s ds}, \quad T > t$$

$$\mathbb{P}\left[\tau \in (s, s+ds) \middle| \tau > t\right] = \lambda_s e^{-\int_t^s \lambda_a da} ds, \quad s > t$$

Derivatives with Default Risk

Consider European payoff $g(X_T)$ at time T where the underlying asset X_t is given by

$$\frac{dX_t}{X_t} = [r(t, X_t) - q(t, X_t)]\, dt + \sigma(t, X_t)\, dW_t.$$

If the counter-party defaults at time τ before T, the contract pays $u_D(\tau, X_\tau)$. The default time τ is assumed to be independent from X. The fair value of this contract at time t, provided that the counter-party has not defaulted yet at time t, is

$$\begin{aligned}
u(t,x) &= \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(s,X_s)ds} g(X_T) 1_{\tau \geq T} + e^{-\int_t^\tau r(s,X_s)ds} u_D(\tau, X_\tau) 1_{\tau < T} \middle| X_t = x, \tau > t \right] \\
&= \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(s,X_s)ds} g(X_T)\, e^{-\int_t^T \lambda_s ds} \middle| X_t = x \right] + \\
&\quad \mathbb{E}^{\mathbb{Q}}\left[\mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^\tau r(s,X_s)ds} u_D(\tau, X_\tau) 1_{\tau < T} \middle| (X_s)_{t \leq s \leq T}, \tau > t \right] \middle| X_t = x, \tau > t \right] \\
&= \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T (r(s,X_s)+\lambda_s)ds} g(X_T) + \int_t^T e^{-\int_t^s (r(\nu,X_\nu)+\lambda_\nu)d\nu} \lambda_s u_D(s, X_s) ds \middle| X_t = x \right]
\end{aligned}$$

Thus, u satisfies the PDE

$$\frac{\partial u}{\partial t} + (r(t,x) - q(t,x))\, x \frac{\partial u}{\partial x} + \frac{1}{2}\sigma^2(t,x) x^2 \frac{\partial^2 u}{\partial x^2} + \lambda_t (u_D(t,x) - u(t,x)) - r(t,x) u = 0$$

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